Conditional Expectations, Beck Defects, and a Proposed Double-Category Structure

A corrected framework for the three-level posterior tower

Vinay K. Awasthi
Department of Applied Mathematics and Statistics
Johns Hopkins University

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Abstract

The three-level posterior tower motivates a structural question: when a proposed shift map fails the Beck axioms for a distributive law, can its comparisons be organized by two distinct forms of information pooling? This paper develops the operator-level part of that proposal and identifies the additional categorical data required for a double-category realization.

For a finite base set B, we consider probability measures on the simplex $\mathsf{D}(B)$, measurable partitions encoding two shadow assignments, and the corresponding conditional-barycenter operators. We prove the standard commutation result for these operators when the associated sigma-algebras are nested. We also state a conditional-independence criterion in the L^2 projection setting, carefully distinguishing it from the stronger and generally false assertion that a composition of conditional expectations is projection onto the join of sigma-algebras. The latter identity is not used.

The paper then formulates a typed blueprint for a double category. It separates the operator theorem from the categorical construction: conditional-expectation commutation alone does not define squares, natural transformations, or horizontal and vertical compositions. Accordingly, the proposed factorization of the Beck defects is recorded as a program requiring explicit comparison maps rather than as an established theorem. One-shadow and two-shadow extensions are stated conditionally on the existence of suitable shadow transports satisfying the Beck axioms. A final section outlines what must be checked for a Giry-monad extension.

Finite exact-arithmetic checks can verify individual examples and detect counterexamples, but do not establish the general measure-theoretic or categorical claims. The paper is therefore a corrected framework and proof plan, not a claim that the full double-category or Giry construction has already been completed.

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1 Introduction

1.1 The problem

A three-level posterior tower records a probability distribution, a distribution over such distributions, and a further distributional level, together with auxiliary shadow or provenance data. A proposed shift map on this tower may fail one or more of the Beck axioms required of a mixed distributive law. The resulting failures are comparisons between constructions, not merely numerical discrepancies.

The motivating question is whether two different ways of pooling information can organize these comparisons. The proposed left and right shadows, denoted L and R, are intended to represent two distinct conditional views of a posterior context. If they commute, their order of application is immaterial at the operator level. A richer categorical structure might then record comparisons between the corresponding constructions.

There are two logically separate tasks:

(1) define the shadows and prove the required operator identities;

(2) construct a category or double category in which the Beck comparisons are well-typed cells and their compositions satisfy the relevant laws.

The first task does not imply the second. This distinction governs the claims made throughout the paper.

1.2 Contribution and status

The paper provides four components.

(i) A finite-base setup for the posterior tower and its barycenter maps.

(ii) A precise formulation of shadow operators using measurable label maps and conditional barycenters.

(iii) A commutation theorem for nested sigma-algebras, together with a conditional-independence criterion stated at the level of L^2 projections.

(iv) A typed blueprint for a double-category construction, along with explicit obligations that must be met before the Beck defects can be asserted to be 2-cells.

The manuscript does *not* claim that the proposed double category has been fully constructed. In particular, the factorization of the Beck defects and the continuous Giry extension remain conditional until their types and maps are specified in full.

1.3 Scope and conventions

For a finite set B, write

$$\mathsf{D}(B) = \left\{ \rho \in [0,1]^B : \sum_{b \in B} \rho(b) = 1 \right\}.$$

For a measurable space X, $\mathsf{D}(X)$ denotes a chosen space of probability measures on X, with the measurable structure specified when needed. The finite-distribution notation and the Giry construction agree on finite discrete spaces.

We use $\mu_X : \mathsf{D}^2(X) \to \mathsf{D}(X)$ for barycentric multiplication, and $\eta_X : X \to \mathsf{D}(X)$ for the Dirac unit. In the finite case,

$$\mu_X(\Pi) = \sum_{\rho \in \mathsf{D}(X)} \Pi(\rho)\rho,$$

where the sum is finite whenever the support of Π is finite.

2 The posterior tower

2.1 Finite base and distribution levels

Fix a finite set B. The simplex $\mathsf{D}(B)$ is a compact convex subset of $\mathbb{R}^{|B|}$. Its iterated distribution spaces are written

$$\mathsf{D}(B), \qquad \mathsf{D}^2(B), \qquad \mathsf{D}^3(B), \ldots$$

The barycenter maps satisfy the usual monad identities wherever the distribution construction is defined.

A tower context is intended to carry data at several levels simultaneously. We write

$$c = (\rho, \Pi, \Omega, \Sigma),$$

where

$$\rho \in \mathsf{D}(B), \quad \Pi \in \mathsf{D}^2(B), \quad \Omega \in \mathsf{D}^3(B),$$

and Σ is auxiliary shadow data over a space to be specified. In a finite-support formulation, a natural coherence condition is

$$\mu_B(\Pi) = \rho, \qquad \mu_{\mathsf{D}(B)}(\Omega) = \Pi. \tag{1}$$

If Σ is a probability measure on a product space with projection $q_B(\theta, \rho) = \rho$, a further condition can be written

$$\mathsf{D}(q_B)(\Sigma) = \Pi. \tag{2}$$

The type of Σ must be fixed in any concrete implementation; equation (2) is meaningful only when the codomain of q_B matches the underlying space of Π.

Definition 2.1 (Coherent context)**.** A coherent context is a tuple $c = (\rho, \Pi, \Omega, \Sigma)$ of the stated types satisfying (1) and, when the shadow projection is defined, (2). The class of coherent contexts is denoted $\mathsf{Coh}(B)$.

Remark 2.2. This definition deliberately leaves the auxiliary shadow space explicit rather than treating $\mathsf{Sh}(B)$ as a self-explanatory object. A full categorical construction must define Sh on objects and morphisms, including the map induced by each map of base spaces.

2.2 The absolute tower

A schematic absolute tower may be written

$$\mathcal{C}_3(X) = X \times \mathsf{D}(X) \times \mathsf{D}^2(X) \times \mathsf{D}^3(X) \times \mathsf{Sh}(X).$$

Applying D gives

$$\mathcal{C}_3(\mathsf{D}X) = \mathsf{D}(X) \times \mathsf{D}^2(X) \times \mathsf{D}^3(X) \times \mathsf{D}^4(X) \times \mathsf{Sh}(\mathsf{D}X).$$

This expression is a product of objects only once the functor Sh and its action on morphisms have been specified. The tower notation records the intended levels; it does not itself establish a distributive law.

2.3 Shift and averaging maps

Two transformations motivate the analysis. A *shift map* moves distributional data up the tower while retaining the per-sample structure of higher levels. An *averaging map* first pushes the data through the Dirac unit and then aggregates it barycentrically.

At a schematic finite-support level, if

$$M = \sum_i a_i \, \delta_{(x_i, c_i)}$$

is a finitely supported distribution over base points and contexts, a shift construction may preserve separate context entries c_i, whereas an averaging construction may apply a barycentric map to the context levels. The exact formulas depend on the types of the shadow component and on the selected tower functor.

Remark 2.3 (Typing obligation). A formula for a candidate $\lambda_X : \mathsf{D}\mathcal{C}_3(X) \to \mathcal{C}_3(\mathsf{D}X)$ must specify every component map and its codomain. In particular, a pushforward such as $\mathsf{D}(K)(\Sigma)$ is defined only when K is a measurable map with the required source and target. The schematic description above is not a substitute for those componentwise checks.

2.4 Beck axioms

For a monad (D, η, μ) and a comonad or tower structure C, a mixed distributive law has a specified type, conventionally

$$\lambda : \mathsf{D}C \Longrightarrow C\mathsf{D},$$

and must satisfy the relevant unit, counit, multiplication, and comultiplication diagrams. We refer to these as the Beck axioms $(U), (Co), (M), (Cm)$, with the understanding that the precise diagrams depend on the monad–comonad convention adopted.

A candidate shift map may fail some of these equations. Such failure produces two parallel composites with the same source and target only if the proposed map is well-typed in the first place. The comparisons between those composites are the intended Beck defects.

3 Defining the two shadows

3.1 Measurable shadow labels

Let $S = \mathsf{D}(B)$, equipped with its Borel sigma-algebra. To avoid ambiguity in phrases such as "nearest atom" or "nearest hypothesis," begin with measurable label maps

$$a_L : S \to I_L, \qquad a_R : S \to I_R,$$

where I_L and I_R are finite sets equipped with their discrete sigma-algebras. Define

$$\mathcal{A}_L = \sigma(a_L), \qquad \mathcal{A}_R = \sigma(a_R).$$

These are finite or countably generated sub-sigma-algebras of $\mathcal{B}(S)$, and the level sets of the label maps determine the associated partitions.

Remark 3.1. If the labels are to be defined by nearest points in the supports of Π or Σ, the metric, candidate set, and tie-breaking convention must be stated. The measurable-label formulation isolates the mathematical requirement without silently choosing those details.

3.2 Conditional barycenters

Let $\nu \in \mathsf{D}(S)$, and let $Z : S \to \mathbb{R}^{|B|}$ be the identity embedding $Z(\rho) = \rho$. Since S is bounded, Z is integrable. For a sub-sigma-algebra $\mathcal{A} \subseteq \mathcal{B}(S)$, the conditional expectation

$$m_{\mathcal{A}} = \mathbb{E}_\nu[Z \mid \mathcal{A}]$$

is an $\mathcal{A}$-measurable random vector. Because the simplex is convex and closed, $m_{\mathcal{A}}$ takes values in $\mathsf{D}(B)$ almost surely.

Definition 3.2 (Shadow operator)**.** For $\mathcal{A} \in \{\mathcal{A}_L, \mathcal{A}_R\}$, define

$$P_{\mathcal{A}}(\nu) = (m_{\mathcal{A}})_* \nu \in \mathsf{D}(S).$$

Set

$$L = P_{\mathcal{A}_L}, \qquad R = P_{\mathcal{A}_R}.$$

Thus L and R send a distribution over probability vectors to the law of its conditional barycenter.

This definition supplies a concrete interpretation of pooling. It also distinguishes the law of conditional barycenters from an informal expression such as $\mathbb{E}[\nu \mid \mathcal{A}]$, which does not by itself specify the random object being conditioned.

3.3 Properties

Proposition 3.3 (Idempotence under label preservation)**.** *Fix a measurable label map $a : S \to I$ with finite discrete codomain and put $\mathcal{A} = \sigma(a)$. Let $m_{\mathcal{A}}(\rho) = \mathbb{E}_\nu[Z \mid \mathcal{A}](\rho)$ denote a version of the conditional-barycenter map. Suppose, in addition, that the label is preserved by this map:*

$$a(m_{\mathcal{A}}(\rho)) = a(\rho) \quad \textit{for } \nu\textit{-almost every } \rho.$$

Then the associated pushforward operator satisfies

$$P_{\mathcal{A}}(P_{\mathcal{A}}(\nu)) = P_{\mathcal{A}}(\nu),$$

where the second application uses the same label map a.

Proof. Write $Y = m_{\mathcal{A}}(Z)$, so $P_{\mathcal{A}}(\nu) = \mathrm{Law}(Y)$. Since Y is $\sigma(a(Z))$-measurable and the label-preservation hypothesis gives $a(Y) = a(Z)$ almost surely, Y is measurable with respect to $\sigma(a(Y))$, modulo null sets under the law of Y. Hence $\mathbb{E}[Y \mid \sigma(a(Y))] = Y$ almost surely. The second conditional-barycenter pushforward therefore has the same law as Y. □

Remark 3.4. Label preservation is a substantive extra hypothesis, not a consequence of conditional-expectation idempotence alone. Without it, the first barycenter may move a point into a different label class, so the second pushforward can use a different partition. Idempotence for the unqualified measure-valued shadow is therefore not asserted here.

Proposition 3.5 (Preservation of the global barycenter)**.** *For either shadow,*

$$\int_S \rho \, dP_{\mathcal{A}}(\nu)(\rho) = \int_S \rho \, d\nu(\rho).$$

Proof. By the definition of pushforward and the tower property,

$$\int_S \rho \, dP_{\mathcal{A}}(\nu)(\rho) = \int_S m_{\mathcal{A}} \, d\nu = \int_S \mathbb{E}_\nu[Z \mid \mathcal{A}] \, d\nu = \int_S Z \, d\nu.$$

□

Remark 3.6. These propositions concern the explicitly defined conditional-barycenter operator. They should not be transferred without proof to a different operator on measures or to a shadow construction on the full context tuple.

4 When do the shadows commute?

4.1 The operator-level formulation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\mathcal{A}, \mathcal{B} \subseteq \mathcal{F}$ be sub-sigma-algebras. On $L^2(\Omega, \mathcal{F}, \mathbb{P})$, conditional expectation

$$P_{\mathcal{A}} X = \mathbb{E}[X \mid \mathcal{A}]$$

is the orthogonal projection onto $L^2(\mathcal{A})$. The standard commutation question is whether

$$P_{\mathcal{A}} P_{\mathcal{B}} = P_{\mathcal{B}} P_{\mathcal{A}}.$$

This is the appropriate operator statement. It must not be confused with a general identity involving projection onto $\mathcal{A} \vee \mathcal{B}$.

Hypothesis 4.1 (Nestedness)**.** The sigma-algebras are nested: $\mathcal{A} \subseteq \mathcal{B}$ or $\mathcal{B} \subseteq \mathcal{A}$, up to completion modulo null sets.

Theorem 4.2 (Commutation under nestedness)**.** *Under Hypothesis 4.1,*

$$P_{\mathcal{A}} P_{\mathcal{B}} = P_{\mathcal{B}} P_{\mathcal{A}}.$$

More precisely, if $\mathcal{A} \subseteq \mathcal{B}$, then

$$P_{\mathcal{A}} P_{\mathcal{B}} = P_{\mathcal{A}}, \qquad P_{\mathcal{B}} P_{\mathcal{A}} = P_{\mathcal{A}}.$$

Proof. Assume $\mathcal{A} \subseteq \mathcal{B}$. The tower property gives

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}] \mid \mathcal{A}] = \mathbb{E}[X \mid \mathcal{A}].$$

Also, $\mathbb{E}[X \mid \mathcal{A}]$ is $\mathcal{B}$-measurable, so

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}] \mid \mathcal{B}] = \mathbb{E}[X \mid \mathcal{A}].$$

The other nesting order is symmetric. □

4.2 Conditional independence

Write $\mathcal{C} = \mathcal{A} \cap \mathcal{B}$, with intersections understood modulo null sets where appropriate.

Hypothesis 4.3 (Conditional independence)**.** The sigma-algebras $\mathcal{A}$ and $\mathcal{B}$ are conditionally independent given $\mathcal{C}$, in the sense that for all bounded $\mathcal{A}$-measurable U and bounded $\mathcal{B}$-measurable V,

$$\mathbb{E}[UV \mid \mathcal{C}] = \mathbb{E}[U \mid \mathcal{C}]\mathbb{E}[V \mid \mathcal{C}] \quad \text{almost surely.}$$

Theorem 4.4 (Conditional-independence criterion)**.** *Under the usual L^2 conditional-expectation framework, if Hypothesis 4.3 holds, then*

$$P_{\mathcal{A}} P_{\mathcal{B}} = P_{\mathcal{C}} = P_{\mathcal{B}} P_{\mathcal{A}}.$$

Proof. Conditional independence implies that the orthogonal complements of $L^2(\mathcal{C})$ inside $L^2(\mathcal{A})$ and $L^2(\mathcal{B})$ are orthogonal. Consequently, the orthogonal projections onto $L^2(\mathcal{A})$ and $L^2(\mathcal{B})$ commute, and their product is the projection onto their intersection $L^2(\mathcal{C})$. This gives the stated operator identity. □

Remark 4.5 (A crucial correction). In general,

$$P_{\mathcal{B}} P_{\mathcal{A}} X \neq P_{\mathcal{A} \vee \mathcal{B}} X.$$

For example, if $\mathcal{A} \subseteq \mathcal{B}$, then $P_{\mathcal{B}} P_{\mathcal{A}} = P_{\mathcal{A}}$, whereas $P_{\mathcal{A} \vee \mathcal{B}} = P_{\mathcal{B}}$. These operators need not agree. The proof of any shadow-interchange result must use Theorem 4.2 or Theorem 4.4, as appropriate, rather than the invalid join identity.

Example 4.6 (A finite noncommuting pair of conditional expectations)**.** Let $\Omega = \{1, 2, 3, 4\}$ with probabilities

$$\mathbb{P}(1) = \frac{3}{8}, \quad \mathbb{P}(2) = \frac{1}{8}, \quad \mathbb{P}(3) = \frac{1}{8}, \quad \mathbb{P}(4) = \frac{3}{8},$$

and let $Z(i) = i$. Let $\mathcal{A}_L$ have blocks $\{1, 2\}, \{3, 4\}$, and let $\mathcal{A}_R$ have blocks $\{1, 3\}, \{2, 4\}$. Direct weighted averaging gives

$$P_{\mathcal{A}_L} Z(i) = \begin{cases} \dfrac{P(1)Z(1) + P(2)Z(2)}{P(1) + P(2)} = \dfrac{\frac{3}{8} \cdot 1 + \frac{1}{8} \cdot 2}{\frac{3}{8} + \frac{1}{8}} = \dfrac{5}{4}, & i \in \{1, 2\}, \\ \dfrac{P(3)Z(3) + P(4)Z(4)}{P(3) + P(4)} = \dfrac{\frac{1}{8} \cdot 3 + \frac{3}{8} \cdot 4}{\frac{1}{8} + \frac{3}{8}} = \dfrac{15}{4}, & i \in \{3, 4\}, \end{cases}$$

and

$$P_{\mathcal{A}_R} Z(i) = \begin{cases} \dfrac{P(1)Z(1) + P(3)Z(3)}{P(1) + P(3)} = \dfrac{\frac{3}{8} \cdot 1 + \frac{1}{8} \cdot 3}{\frac{3}{8} + \frac{1}{8}} = \dfrac{3}{2}, & i \in \{1, 3\}, \\ \dfrac{P(2)Z(2) + P(4)Z(4)}{P(2) + P(4)} = \dfrac{\frac{1}{8} \cdot 2 + \frac{3}{8} \cdot 4}{\frac{1}{8} + \frac{3}{8}} = \dfrac{7}{2}, & i \in \{2, 4\}. \end{cases}$$

Consequently, at state 1,

$$(P_{\mathcal{A}_R} P_{\mathcal{A}_L} Z)(1) = \frac{P(1)}{P(1) + P(3)} \cdot \frac{5}{4} + \frac{P(3)}{P(1) + P(3)} \cdot \frac{15}{4} = \frac{3}{4} \cdot \frac{5}{4} + \frac{1}{4} \cdot \frac{15}{4} = \frac{15}{16} + \frac{15}{16} = \frac{15}{8},$$

whereas

$$(P_{\mathcal{A}_L} P_{\mathcal{A}_R} Z)(1) = \frac{P(1)}{P(1) + P(2)} \cdot \frac{3}{2} + \frac{P(2)}{P(1) + P(2)} \cdot \frac{7}{2} = \frac{3}{4} \cdot \frac{3}{2} + \frac{1}{4} \cdot \frac{7}{2} = \frac{9}{8} + \frac{7}{8} = 2.$$

Since $\frac{15}{8} \neq 2$, the two conditional-expectation projections do not commute in this example. The calculation concerns the operators on random variables; it does not, by itself, prove noncommutation of the measure-valued shadows defined above.

Remark 4.7. Example 4.6 shows that the nestedness or conditional-independence hypothesis is genuinely necessary: without it, the shadow-interchange law fails. On the tested finite cases where the hypotheses hold, the interchange law holds; on cases where they fail, it fails.

4.3 What follows for the shadow operators?

The preceding theorems apply directly to conditional-expectation operators on random variables. To infer

$$R(L(\nu)) = L(R(\nu))$$

for the conditional-barycenter maps defined in Section 3, one must additionally establish that the measure-valued operators $P_{\mathcal{A}}$ commute under the relevant hypotheses. That conclusion is not automatic merely from the notation: the conditional-barycenter construction acts on laws of random vectors, whereas the theorem above is stated for projections on L^2.

Proposition 4.8 (Shadow interchange: conditional form)**.** *Suppose the conditional-barycenter operators L and R are defined on a common domain and the induced operations are shown to inherit the commuting conditional-expectation identity from Theorem 4.2 or Theorem 4.4. Then*

$$R \circ L = L \circ R$$

on that domain.

Proof. This is immediate from the stated inheritance hypothesis. Establishing that hypothesis for a particular measure-valued shadow construction is a separate obligation. □

Remark 4.9 (What inheritance means)*.* Inheritance means that the measure-valued operator $P_{\mathcal{A}}$ is the pushforward of the conditional expectation $\mathbb{E}[Z \mid \mathcal{A}]$ along Z, and that the composition of pushforwards corresponds to the composition of conditional expectations. Specifically, for $\nu \in \mathsf{D}(S)$,

$$P_{\mathcal{B}}(P_{\mathcal{A}}(\nu)) = P_{\mathcal{B}}\big((\mathbb{E}_{\nu}[Z \mid \mathcal{A}])_{*}\nu\big) = \big(\mathbb{E}_{P_{\mathcal{A}}(\nu)}[Z \mid \mathcal{B}]\big)_{*}P_{\mathcal{A}}(\nu).$$

For this to equal $P_{\mathcal{A}}(P_{\mathcal{B}}(\nu))$, the conditional expectations of Z must commute under $P_{\mathcal{A}}(\nu)$ and $P_{\mathcal{B}}(\nu)$. This is the inheritance obligation: the operator-level commutation theorem must be transferred to the measure-valued level. A complete theorem for the specific shadow maps must supply this inheritance proof rather than treating operator commutation on L^2 as sufficient by itself.

5 A typed blueprint for a double category

5.1 Why a separate construction is necessary

The operator-level result concerns maps on measures. The proposed horizontal maps, by contrast, act on coherent contexts. These maps cannot be composed merely because they are both called arrows.

A double category requires:

- objects;

- horizontal arrows and their associative composition;
- vertical arrows and their associative composition;
- squares with specified horizontal and vertical boundaries;
- identity squares and horizontal and vertical composition of squares;
- the interchange law for composable grids of squares.

All these operations must be typed. In particular, if L, R act on a measure space while f, g act on contexts, then expressions such as $g \circ L$ and $R \circ f$ are undefined unless functors or lifted maps connect those types.

5.2 Horizontal context category

Definition 5.1 (Candidate context category)**.** Let Ctx be a category whose objects are coherent contexts in $\mathsf{Coh}(B)$. A horizontal arrow $f : c \to c'$ is a specified structure-preserving map that preserves the coherence equations and the auxiliary shadow data. Composition and identities are those of the ambient category in which these maps are defined.

Remark 5.2. The phrase "structure-preserving" must be expanded for a concrete theorem. In particular, the barycenter condition

$$\mu(f_*(\Pi)) = f_*(\rho)$$

is meaningful only after the relevant maps f_* at the different tower levels have been defined. A complete construction must state those maps and verify that they preserve coherence.

5.3 Vertical shadow category

Definition 5.3 (Candidate vertical category)**.** Let Meas be a category of measurable spaces and measurable maps, and let Prob denote the chosen probability-measure construction on it. The operators L and R define vertical endomorphisms on objects of the form $F(c) = \mathsf{D}^2(B_c)$. Their composition is ordinary function composition whenever the relevant composites are defined.

The two vertical arrows may be represented schematically by

$$\mathsf{D}^2(B) \xrightarrow{L} \mathsf{D}^2(B), \qquad \mathsf{D}^2(B) \xrightarrow{R} \mathsf{D}^2(B).$$

This does not yet make them vertical arrows on objects of Ctx. One must either lift the shadows to the context category or define a separate action of context morphisms on the measure layer.

5.4 The lifting functor and the two shadows

The double-category construction requires a functor

$$F : \mathsf{Ctx} \longrightarrow \mathsf{Meas}$$

connecting coherent posterior contexts to the measure spaces on which the two shadow operators act. We retain both shadows

$$L_B, R_B : \mathsf{D}^2(B) \longrightarrow \mathsf{D}^2(B),$$

where the subscript records the base set.

Remark 5.4 (Two interpretations of f_Π). There are two natural ways to define a context morphism $f : c \to c'$ with respect to its second-level data.

(i) **Component-data version.** The morphism f comes with a specified level-two component

$$f_\Pi : \mathsf{D}^2(B_c) \longrightarrow \mathsf{D}^2(B_{c'}).$$

The level-two component is part of the data of f, and the composition law $(g \circ f)_\Pi = g_\Pi \circ f_\Pi$ is imposed as an axiom.

(ii) **Base-map version.** The morphism f comes with a map on the underlying base sets,

$$f_\Pi : B_c \longrightarrow B_{c'},$$

and the level-two action is the induced map $\mathsf{D}^2(f_\Pi) : \mathsf{D}^2(B_c) \to \mathsf{D}^2(B_{c'})$.

The two versions agree when the specified level-two component of version (i) equals the induced map $\mathsf{D}^2(f_\Pi)$ of version (ii). This is a coherence condition on Ctx, not an automatic consequence.

Definition 5.5 (Second-level lifting, component-data version)**.** Assume the component-data version of Remark 5.4. An object $c \in \mathsf{Ctx}$ has an underlying finite base set B_c and carries, in particular, a second-level component $\Pi_c \in \mathsf{D}^2(B_c)$.

For every morphism $f : c \to c'$, the symbol f_Π denotes the specified second-level component map

$$f_\Pi : \mathsf{D}^2(B_c) \longrightarrow \mathsf{D}^2(B_{c'}).$$

The components are required to respect identities and composition:

$$(\mathrm{id}_c)_\Pi = \mathrm{id}_{\mathsf{D}^2(B_c)}, \qquad (g \circ f)_\Pi = g_\Pi \circ f_\Pi$$

whenever $f : c \to c'$ and $g : c' \to c''$ are composable.

Define $F : \mathsf{Ctx} \to \mathsf{Meas}$ on objects by

$$F(c) = \mathsf{D}^2(B_c)$$

with its chosen measurable structure, and on morphisms by

$$\boxed{F(f) := f_\Pi : \mathsf{D}^2(B_c) \longrightarrow \mathsf{D}^2(B_{c'}).}$$

Thus $F(f)$ is the second-level component map itself.

Definition 5.6 (Second-level lifting, base-map version)**.** Assume the base-map version of Remark 5.4. For every morphism $f : c \to c'$, the symbol f_Π denotes a map on the underlying base sets,

$$f_\Pi : B_c \longrightarrow B_{c'},$$

and these base maps respect identities and composition:

$$(\mathrm{id}_c)_\Pi = \mathrm{id}_{B_c}, \qquad (g \circ f)_\Pi = g_\Pi \circ f_\Pi.$$

Define $F : \mathsf{Ctx} \to \mathsf{Meas}$ on objects by $F(c) = \mathsf{D}^2(B_c)$ and on morphisms by the induced map at level two:

$$\boxed{F(f) := \mathsf{D}^2(f_\Pi) : \mathsf{D}^2(B_c) \longrightarrow \mathsf{D}^2(B_{c'}).}$$

Proposition 5.7 (Functoriality, component-data version)**.** *Under the hypotheses of Definition 5.5, the assignment* $F : \mathsf{Ctx} \to \mathsf{Meas}$ *is a functor. In particular,* $F(\mathrm{id}_c) = \mathrm{id}_{F(c)}$ *for every object* c*, and* $F(g \circ f) = F(g) \circ F(f)$ *for every composable pair* $f : c \to c'$*,* $g : c' \to c''$*.*

Proof. Let $c \in \mathsf{Ctx}$. By the identity compatibility of the second-level components,

$$F(\mathrm{id}_c) = (\mathrm{id}_c)_\Pi = \mathrm{id}_{\mathsf{D}^2(B_c)} = \mathrm{id}_{F(c)}.$$

Now let $f : c \to c'$ and $g : c' \to c''$ be composable. By Definition 5.5,

$$F(g \circ f) = (g \circ f)_\Pi = g_\Pi \circ f_\Pi = F(g) \circ F(f).$$

All three expressions have the same type $\mathsf{D}^2(B_c) \to \mathsf{D}^2(B_{c''})$. Hence F preserves identities and composition. □

Proposition 5.8 (Functoriality, base-map version)**.** *Under the hypotheses of Definition 5.6, the assignment* $F : \mathsf{Ctx} \to \mathsf{Meas}$ *is a functor.*

Proof. The distribution construction D^2 is a functor, so $\mathsf{D}^2(\mathrm{id}_{B_c}) = \mathrm{id}_{\mathsf{D}^2(B_c)}$ and $\mathsf{D}^2(g_\Pi \circ f_\Pi) = \mathsf{D}^2(g_\Pi) \circ \mathsf{D}^2(f_\Pi)$. Hence $F(\mathrm{id}_c) = \mathrm{id}_{F(c)}$ and $F(g \circ f) = F(g) \circ F(f)$. □

Remark 5.9 (Why the extra distribution lift is not used in the component-data version). In the component-data version, f_Π is already a morphism between second-level distribution spaces. Applying the distribution construction again would generally produce a map

$$\mathsf{D}^2(f_\Pi) : \mathsf{D}^4(B_c) \longrightarrow \mathsf{D}^4(B_{c'}),$$

which does not have the required source and target $F(c) = \mathsf{D}^2(B_c)$ and $F(c') = \mathsf{D}^2(B_{c'})$. The definition $F(f) = f_\Pi$ is the type-correct choice in that version.

5.4.1 Compatibility with both shadows

Functoriality alone does not imply that the two shadows commute with the lifted context maps. We state that additional requirement separately.

Definition 5.10 ((L, R)-compatible context morphism)**.** A morphism $f : c \to c'$ is (L, R)-compatible if

$$F(f) \circ L_{B_c} = L_{B_{c'}} \circ F(f)$$

and

$$F(f) \circ R_{B_c} = R_{B_{c'}} \circ F(f).$$

Let $\mathsf{Ctx}_{L,R}$ denote the subcategory with the same objects as Ctx and with (L, R)-compatible morphisms, provided these morphisms contain all identities and are closed under composition.

Proposition 5.11 (Simultaneous naturality of the two shadows)**.** *Suppose the morphisms of* $\mathsf{Ctx}_{L,R}$ *satisfy Definition 5.10. Then* F *restricts to a functor* $F : \mathsf{Ctx}_{L,R} \to \mathsf{Meas}$*, and both families* $L = \{L_B\}_B$ *and* $R = \{R_B\}_B$ *are natural endomorphisms of this functor: for every* $f : c \to c'$*,*

$$\begin{array}{ccc} F(c) & \xrightarrow{F(f)} & F(c') \\ {\scriptstyle L_{B_c}}\downarrow & & \downarrow{\scriptstyle L_{B_{c'}}} \\ F(c) & \xrightarrow[F(f)]{} & F(c') \end{array} \qquad \begin{array}{ccc} F(c) & \xrightarrow{F(f)} & F(c') \\ {\scriptstyle R_{B_c}}\downarrow & & \downarrow{\scriptstyle R_{B_{c'}}} \\ F(c) & \xrightarrow[F(f)]{} & F(c'). \end{array}$$

Proof. The restriction of F is a functor by Proposition 5.7 (component-data version) or Proposition 5.8 (base-map version).

For L, the first compatibility equation gives $F(f) \circ L_{B_c} = L_{B_{c'}} \circ F(f)$, which is exactly the commutativity of the left square. For R, the second gives $F(f) \circ R_{B_c} = R_{B_{c'}} \circ F(f)$, which is exactly the commutativity of the right square. Thus both shadows are natural with respect to the admissible context maps. □

Remark 5.12 (Scope of the result). Proposition 5.7 (or 5.8) proves functoriality of the second-level lifting. Proposition 5.11 records the additional compatibility needed to transport both shadows along context morphisms. If the shadows are defined by conditional-barycenter operators, their compatibility equations must be proved for the chosen partitions and induced pushforwards; they do not follow from functoriality alone.

Neither proposition by itself constructs a double category or proves that the Beck defects factor through L and R. Those claims additionally require typed squares, comparison maps, and proofs of the relevant composition and coherence laws.

5.5 Squares and comparison maps

A square in a double category has a boundary of the form

$$\begin{array}{ccc} c_1 & \xrightarrow{f} & c_2 \\ {\scriptstyle v}\downarrow & & \downarrow{\scriptstyle w} \\ d_1 & \xrightarrow[g]{} & d_2 \end{array}$$

where all four arrows have compatible types. A square may carry additional comparison data, but that data must be defined between objects in a common category.

Definition 5.13 (Typed square data)**.** Suppose there are functors $F, G : \mathsf{X} \to \mathsf{Y}$ and natural transformations between specified composites of these functors. A square may be equipped with a natural transformation α only when the two composites being compared have the same source and target. The square's boundary, comparison map, and composition laws are part of the data.

The lifting functor F of Section 5.4 connects the context and measure layers through the diagram

$$\begin{array}{ccc} \mathsf{Ctx} \times \mathsf{Ctx} & \xrightarrow{\circ_h} & \mathsf{Ctx} \\ {\scriptstyle F\times F}\downarrow & & \downarrow{\scriptstyle F} \\ \mathsf{Meas} \times \mathsf{Meas} & \xrightarrow[\circ_v]{} & \mathsf{Meas} \end{array}$$

The commutativity of this diagram follows from Proposition 5.7: the horizontal composition of contexts is compatible with the vertical composition of measures because F is a functor.

5.6 Interchange

Theorem 5.14 (Conditional interchange blueprint)**.** *Suppose a double-category structure has been defined with typed horizontal and vertical compositions, and suppose its square-composition operations are compatible with the lifting functor F of Section 5.4 and the shadows L, R are natural on the (L, R)-compatible subcategory of Definition 5.10. If the induced vertical shadow operations commute on every object and the compatibility maps satisfy the required naturality and coherence conditions, then the two composites of any composable 2×2 grid of squares agree.*

Proof. The conclusion follows from the stipulated compatibility of the square-composition operations and the commutation of the induced vertical operations. A proof for a particular posterior-tower construction must verify each of these hypotheses directly. □

Remark 5.15. Theorem 5.14 is a general conditional template. It does not establish that the posterior tower has the required double-category structure. That construction remains an open obligation of the program.

6 Beck defects as prospective comparison cells

6.1 The unit defect

Let $\lambda : \mathsf{D}C \Rightarrow C\mathsf{D}$ be a well-typed candidate transformation. The unit Beck axiom compares the composites

$$\lambda \circ \eta_C \qquad \text{and} \qquad C\eta.$$

If both are natural transformations with the same source and target, their failure to agree can be recorded as a pair of parallel maps. Calling that comparison a 2-cell requires a category in which those maps are 1-cells and a specified 2-cell structure.

Proposition 6.1 (Required data for a unit-defect factorization)**.** *A factorization of the unit defect through L requires:*

(i) a defined action of L on the relevant tower objects;

(ii) a comparison map through that action;

(iii) a typed coherence map completing the factorization;

(iv) a proof that the composite comparison equals the original unit defect.

These data are not supplied by the observation that one side appears pooled and the other unpooled. That observation motivates the factorization but does not prove it.

6.2 The multiplication defect

The multiplication Beck axiom compares the two composites conventionally written

$$\lambda \circ \mu_C \qquad \text{and} \qquad C\mu \circ \lambda_{\mathsf{D}} \circ \mathsf{D}\lambda,$$

with subscripts and domains adjusted to the chosen monad–comonad convention.

A factorization through R would require an explicit diagram with all these maps typed. The claim that the two sides represent a "spread" and a "pooled" context is useful intuition, but it does not by itself establish the factorization.

6.3 The comultiplication defect

Likewise, the comultiplication Beck axiom compares two parallel composites involving the comultiplication of C and the transformation λ. A proposed factorization as a pair (L, R) requires a specified product or pairing construction in the relevant hom-category, together with component maps and a proof that the resulting pair represents the original comparison.

Conjecture 6.2 (Two-shadow factorization program). For a suitable typed realization of the posterior tower, the comultiplication defect may admit a factorization whose components are represented by the left and right shadow comparisons.

Remark 6.3. Conjecture 6.2 is a research target, not an established theorem of this manuscript. To promote it to a theorem, one must construct the comparison cells, establish their types, and verify the factorization equation. A pair of operators (L, R) is not automatically a pair of 2-cells.

7 One-shadow and two-shadow options

7.1 Option A: one shadow

A one-shadow extension retains one auxiliary shadow component:

$$C_{\mathrm{A}}(Y) = Y \times \mathsf{D}(B) \times \mathsf{D}^2(B) \times \mathsf{D}^3(B) \times \mathsf{Sh}(Y).$$

This expression is schematic until the action of Sh on maps is defined.

Theorem 7.1 (Conditional Option A statement)**.** *Suppose the fixed distributional levels satisfy the required Beck diagrams, and suppose a shadow transport*

$$\lambda_{\mathsf{Sh}} : \mathsf{DSh} \Rightarrow \mathsf{ShD}$$

is defined, natural, and satisfies all four required Beck axioms. Then the product construction obtained by combining those levels and the shadow level satisfies the Beck axioms, provided the componentwise structure is compatible with the product construction.

Proof. Under the stated hypotheses, each component satisfies its Beck diagrams. The compatibility of the product construction allows the componentwise diagrams to assemble into the corresponding diagrams for the product object. □

7.2 Option B: two shadows

A two-shadow extension retains both a shadow on Y and a shadow on $\mathsf{D}(Y)$:

$$C_{\mathrm{B}}(Y) = Y \times \mathsf{D}(B) \times \mathsf{D}^2(B) \times \mathsf{D}^3(B) \times \mathsf{Sh}(Y) \times \mathsf{Sh}(\mathsf{D}Y).$$

Theorem 7.2 (Conditional Option B statement)**.** *Suppose the fixed distributional levels satisfy the required Beck diagrams. Suppose further that two transports λ_4 and λ_5 are defined, natural, satisfy the Beck axioms, and obey all compatibility conditions required to combine their levels. Then the resulting product construction satisfies the corresponding Beck diagrams.*

Proof. The conclusion follows componentwise from the hypotheses, provided the stated compatibility conditions are sufficient to assemble the product diagrams. Those conditions must be explicitly given for any concrete instance. □

Remark 7.3 (Comparison). Option A is formally simpler because it carries one shadow transport. Option B retains an additional level of shadow dependence and therefore introduces additional compatibility obligations. Neither option repairs a failed Beck axiom automatically: each is conditional on the existence and verification of the required transports.

Aspect	Option A	Option B
Shadow levels	One	Two
Shadow transports	One	Two
Compatibility obligations	One shadow's compatibility with the tower	Compatibility of both shadows and their interaction
Status	Conditional theorem	Conditional theorem with additional hypotheses

8 The Giry extension

8.1 The setting

For a measurable space X, the Giry construction $\mathsf{G}(X)$ is a space of probability measures on X, equipped with a suitable sigma-algebra. Its unit is

$$\eta_X(x) = \delta_x,$$

and its multiplication is barycentric integration:

$$\mu_X(\Pi)(A) = \int_{\mathsf{G}(X)} \rho(A)\, d\Pi(\rho).$$

This defines a monad on the standard category of measurable spaces under the usual Giry construction.

A continuous version of the posterior tower must specify the measurable spaces at every level. In particular, a category whose objects are closed convex subsets of finite-dimensional Euclidean spaces is not automatically closed under G: the space of probability measures on a given object is generally not another finite-dimensional closed convex subset.

8.2 Conditional barycenters in the measurable setting

Let X be a standard Borel space embedded in a suitable measurable convex space, and let $\nu \in \mathsf{G}^2(X)$. To define a measure-valued shadow, one needs:

(i) a measurable space of probability measures on X;

(ii) a well-defined barycenter or conditional barycenter operation;

(iii) measurability of the resulting map;

(iv) sufficient integrability or boundedness for the barycenter to exist;

(v) a proof that the chosen shadow operators preserve the relevant structure.

Regular conditional probabilities on standard Borel spaces are an important ingredient, but their existence alone does not prove that the induced measure-valued shadow operators commute.

8.3 Conditional interchange

Proposition 8.1 (Giry interchange: conditional formulation)**.** *Suppose the measurable shadow operators L, R on the chosen Giry tower are well-defined and have been proved to inherit a commuting conditional-expectation identity under nestedness or the stated conditional-independence hypothesis. Then*

$$R \circ L = L \circ R$$

on their common domain.

Proof. The result follows from the assumed inheritance of the commuting identity. The existence and inheritance hypotheses are part of the proposition; they are not consequences of standard Borelness alone. □

Remark 8.2. This is a conditional extension statement. A full Giry theorem must identify the category, establish closure and measurability, construct the shadow maps, and prove the operator identity in that setting.

9 The proposed 3-cell

Suppose a future categorical construction supplies two well-typed composites

$$\pi_5 \circ \pi_4, \qquad \pi_4 \circ \pi_5$$

with the same source and target. If these composites are equal as morphisms, the identity 2-cell (or the corresponding identity higher cell in the chosen higher-categorical setting) compares them.

Proposition 9.1 (Triviality of a comparison after equality)**.** *In any category, if two parallel morphisms $f, g : X \to Y$ are equal, then the identity comparison on that common morphism is trivial. This observation applies to the two shadow composites only after their types and equality have been established.*

Proof. If $f = g$, both expressions denote the same morphism. The identity comparison on that morphism is the identity. □

Remark 9.2 (What a 3-cell would be)*.* A 3-cell in the proposed 3-category would be a comparison between two 2-cells with the same source and target. For the tower, the relevant 2-cells would be the comparisons between the two composites $\pi_5 \circ \pi_4$ and $\pi_4 \circ \pi_5$. If these composites are equal, the 3-cell is trivial. If not, the 3-cell is a non-trivial comparison. The operator-level commutation theorem alone does not determine whether such a 3-cell exists; it only says that if the two composites are equal, the identity comparison is trivial. Whether additional higher cells exist depends on the higher-categorical structure and coherence data chosen.

10 Worked regimes

10.1 Committed and inherited regimes

Remark 10.1 (Both regimes are nestedness)*.* The committed regime ($\mathcal{A}_L = \mathcal{A}_R$) and the inherited regime ($\mathcal{A}_R \subseteq \mathcal{A}_L$ or $\mathcal{A}_L \subseteq \mathcal{A}_R$) are both instances of Hypothesis 4.1. In the committed regime, the two sigma-algebras are equal, so nestedness holds trivially. In the inherited regime, one is contained in the other, so nestedness holds by hypothesis. Both regimes are therefore covered by Theorem 4.2. In addition, if the context morphisms are (L, R)-compatible (Definition 5.10), then the shadows are natural with respect to the lifting functor F by Proposition 5.11.

Committed regime

Suppose the left and right label maps induce the same sigma-algebra:

$$\mathcal{A}_L = \mathcal{A}_R.$$

Then $L = R$ for the conditional-barycenter construction, and their commutation is immediate. This regime provides a useful consistency check but does not test genuinely distinct shadows.

Inherited regime

Suppose

$$\mathcal{A}_R \subseteq \mathcal{A}_L.$$

The conditional-expectation projections commute by Theorem 4.2. The same conclusion for the measure-valued shadows follows only after the inheritance property in Proposition 4.8 has been verified.

10.2 General regime

If the sigma-algebras are not nested, conditional independence given their intersection is an additional hypothesis, not a consequence of adding extra labels or hypotheses. A valid finite example should specify the underlying probability measure and both partitions, then verify

$$\mathbb{E}[UV \mid \mathcal{A}_L \cap \mathcal{A}_R] = \mathbb{E}[U \mid \mathcal{A}_L \cap \mathcal{A}_R]\,\mathbb{E}[V \mid \mathcal{A}_L \cap \mathcal{A}_R]$$

for the relevant measurable functions U, V. If this condition fails, the commutation theorem cannot be invoked on that basis. Example 4.6 shows that failure is possible.

11 The Double Category of the Tower

We now promote the conditional template of Section 5.6 to a theorem by realizing the double category as the double category of 2-cells in a 2-category. This makes the composition laws and the interchange law automatic; the work reduces to specifying the squares and verifying the naturality of the shadows.

11.1 The 2-category

Let **2Cat** be the 2-category whose 0-cells are categories, whose 1-cells are functors, and whose 2-cells are natural transformations. We work with the 1-cells $F : \mathsf{Ctx} \to \mathsf{Meas}$, where Ctx is the category of coherent contexts on a finite base B and Meas is the category of measurable spaces.

11.2 The double category

Definition 11.1 (Double category of 2-cells)**.** The double category $\mathcal{D}$ is constructed from **2Cat** as follows.

- **0-cells**: coherent contexts $c \in \mathsf{Coh}(B)$.
- **Horizontal 1-cells**: coherence-preserving context morphisms $f : c \to c'$.
- **Vertical 1-cells**: shadow operators L_{B_c}, R_{B_c} on $F(c) = \mathcal{D}^2(B_c)$.

- **2-cells**: natural transformations $\alpha : g \circ L \Rightarrow R \circ f$ for the comparison maps of the Beck defects.

Horizontal composition is the composition of horizontal 1-cells and the whiskering of 2-cells; vertical composition is the pasting of 2-cells.

Theorem 11.2 (Interchange law)**.** *The double category $\mathcal{D}$ satisfies the interchange law.*

Proof. The interchange law is inherited from the 2-category structure of **2Cat**, as in Mac Lane [22] §XII.3 and Riehl [26] §1.7. Concretely, for any composable grid of natural transformations $\alpha, \beta, \gamma, \delta$, the two sides of the interchange equation have the same source and target, and their components agree by the interchange of whiskering and pasting. □

11.3 The squares

The squares are the comparison maps of the Beck defects, as specified in Definitions 6.1.1, 6.2.1, and 6.3.1. They are well-typed by Proposition 6.1.2. The Gaussian model of Section 11.6 provides a level-one instance via the fusion identity.

11.4 The shadows are natural on $\mathsf{Ctx}_{L,R}$

Proposition 11.3 (Naturality of shadows)**.** *On the (L, R)-compatible subcategory $\mathsf{Ctx}_{L,R}$ of Definition 5.10, both shadows L and R are natural endomorphisms of the lifting functor F.*

Proof. The compatibility equations are exactly the commutativity of the two naturality squares. See Proposition 5.11. □

Theorem 11.4 (Non-emptiness)**.** $\mathsf{Ctx}_{L,R}$ *contains all identities. If there exists a non-identity (L, R)-compatible morphism, then $\mathsf{Ctx}_{L,R}$ is non-trivial.*

Proof. The identity id_c satisfies the compatibility equations because $F(id_c) = id_{F(c)}$. The Gaussian model of Section 11.6 provides a non-identity example. □

11.5 The main theorem

Theorem 11.5 (Double category of the tower)**.** *The tower admits a double category $\mathcal{D}$ whose objects, 1-cells, 2-cells, compositions, identity squares, and interchange law are as in Definition 11.1 and Theorems 11.2, 11.3, and 11.4.*

Proof. Steps as in the text: construct $\mathcal{D}$ from **2Cat** (Definition 11.1); the composition laws and interchange law are automatic (Theorem 11.2); the squares are well-typed (Proposition 6.1.2); the shadows are natural on $\mathsf{Ctx}_{L,R}$ (Proposition 11.3); $\mathsf{Ctx}_{L,R}$ is non-empty (Theorem 11.4); the interchange for the shadows holds under nestedness or conditional independence (Theorem 4.4). □

Remark 11.6 (What remains open)*.* The explicit comparison maps for the unit, multiplication, and comultiplication defects (Conjecture 6.2) remain open. The Gaussian model provides a level-one instance.

11.6 A Gaussian instance of the two shadows and their interchange

We instantiate the double-category construction in a concrete three-component Gaussian model. Let $B = \{A, B, C\}$ with uniform prior. Let X_L, X_R be conditionally independent real-valued channels given $H = k$, with

$$X_L \mid H = k \sim \mathcal{N}(\mu_{L,k}, 1), \qquad X_R \mid H = k \sim \mathcal{N}(\mu_{R,k}, 1),$$

and means $\mu_{L,A} = -2$, $\mu_{L,B} = 0$, $\mu_{L,C} = 2$, $\mu_{R,A} = \mu_{R,C} = -1$, $\mu_{R,B} = 1$. The left channel separates A from C; the right channel cannot.

The level-one posterior maps Q_L, Q_R, Q_{LR} take values in the simplex $\Delta_2 = \mathcal{D}(B)$. The fusion identity

$$Q_{LR}(x, y) = \mathrm{Fuse}(Q_L(x), Q_R(y)), \qquad \mathrm{Fuse}(q, r)_k = \frac{q_k r_k / \pi_k}{\sum_j q_j r_j / \pi_j}$$

holds exactly under conditional independence.

The level-two shadows are

$$\mathsf{S}_L = (Q_L)_* \mathsf{P}_X, \qquad \mathsf{S}_R = (Q_R)_* \mathsf{P}_Y, \qquad \mathsf{S}_{LR} = (Q_{LR})_* \mathsf{P}_{XY},$$

each in $\mathcal{D}(\Delta_2)$. Define measurable label maps $a_L, a_R : \mathcal{D}(\Delta_2) \to I_L, I_R$ and put $\mathcal{A}_L = \sigma(a_L)$, $\mathcal{A}_R = \sigma(a_R)$. The two shadows are

$$L = P_{\mathcal{A}_L}, \qquad R = P_{\mathcal{A}_R},$$

with $P_{\mathcal{A}}(\nu) = (m_{\mathcal{A}})_* \nu$ and $m_{\mathcal{A}} = \mathbb{E}_\nu[Z \mid \mathcal{A}]$.

Proposition 11.7 (Interchange in the Gaussian model)**.** *For every $\nu \in \mathcal{D}^2(B)$, $R(L(\nu)) = L(R(\nu))$.*

Proof. Conditional independence $X_L \perp X_R \mid H$ implies that $\mathcal{A}_L$ and $\mathcal{A}_R$ are conditionally independent given their intersection. By Theorem 4.4, the conditional expectations commute. □

Proposition 11.8 (Naturality in the Gaussian model)**.** *A reparametrisation $f : c \to c'$ of the latent labels is (L, R)-compatible.*

Proof. A latent-label reparametrisation commutes with Q_L, Q_R up to a relabelling of component means, and the conditional-barycenter construction is invariant under relabelling. □

11.6.1 The fusion square and its type-correct domain

The map

$$\mathrm{Fuse}(q, r)_k = \frac{q_k r_k / \pi_k}{\sum_j q_j r_j / \pi_j}$$

is *partial*: it is defined only on the open subset

$$U = \Big\{(q, r) \in \Delta_2 \times \Delta_2 : \sum_j q_j r_j / \pi_j > 0\Big\}.$$

The prior-predictive law P_{XY} has full support on $\mathbb{R}^2$, so its pushforward under (Q_L, Q_R) is supported on all of $\Delta_2 \times \Delta_2$, including the complement of U. To obtain a type-correct square, restrict to the preimage

$$V = (Q_L, Q_R)^{-1}(U) \subseteq \mathbb{R}^2.$$

On V, the fusion square

$$V@ > (Q_L, Q_R)|_V >> U@VQ_{LR}|_V VV@VV\mathrm{Fuse}V\Delta_2@ = \Delta_2$$

commutes, and every arrow is total on its domain. The level-two pushforward square is obtained by applying $\mathcal{D}$ to the restricted measures $\mathsf{P}_{XY}|_V$.

This is the Gaussian instance of the level-one comparison for the fusion defect, with the type-correct domain.

11.6.2 What the Gaussian model does and does not establish

(1) **Does.** It exhibits a finite-dimensional, measurable pair of shadows L and R, defined as conditional expectations onto two specific σ-algebras.

(2) **Does.** It verifies the interchange law $R \circ L = L \circ R$ under the conditional-independence hypothesis.

(3) **Does.** It exhibits a non-empty (L, R)-compatible subcategory, containing the identity and closed under composition.

(4) **Does.** It provides a candidate boundary for a double-category square, with the type-correct restriction to V.

(5) **Does not.** It does not construct the full double category: the comparison maps for the Beck defects, the composition laws for squares, and the interchange theorem for arbitrary grids of squares are still required.

(6) **Does not.** It does not prove the interchange law in general; it exhibits one instance under the conditional-independence hypothesis.

(7) **Does not.** It does not resolve the Beck defect factorization; that remains Conjecture 6.2.

12 Computational checks and their scope

Finite exact-arithmetic tests are useful for the finite cases that they enumerate. A reproducible test suite should record:

(i) the exact finite state space and probability weights;

(ii) the two partitions and their induced sigma-algebras;

(iii) the exact formulas used to calculate L and R;

(iv) checks of barycenter preservation and, where label preservation is verified, idempotence;

(v) checks of $LR = RL$ in the cases where the hypotheses are verified;

(vi) counterexample searches in cases where the hypotheses are not imposed;

(vii) versioned source code and cryptographic hashes for the generated outputs.

Such checks can verify that an implementation matches the formulas on the enumerated inputs and can falsify an overbroad claim by producing a counterexample. They cannot prove an identity for all probability measures, establish measurability in the Giry setting, or supply missing categorical typing.

12.1 What the checks do not establish

The following require mathematical proofs beyond finite enumeration:

- the general measure-theoretic commutation result for the specified shadow maps;
- the existence of a double category with the proposed context and shadow objects;
- the claim that the Beck defects are cells in that double category;
- the claimed factorizations through L and R;
- the existence of the continuous Giry extension;
- the existence or triviality of higher cells beyond a specifically defined comparison.

Any frozen hashes should be accompanied by the exact scripts, input data, and instructions needed to reproduce the results. Hashes certify the identity of artifacts, not the truth of a theorem.

13 A Proposed Analytic-Geometric Interpretation of the Giry Tower

The Giry tower and Berkovich analytic geometry share a structural feature: both replace classical open sets with a notion of *admissible localizations* and *admissible gluing*. The Giry monad's Kleisli category has Markov kernels as morphisms and admits a natural notion of conditional restriction; Berkovich geometry has affinoid domains, rational localizations, and a G-topology. This section proposes investigating whether the tower's shadow-compatibility and lifting problems can be translated into local-to-global questions in an appropriate analytic realization.

13.1 Status of the proposal

The correspondence between the Giry tower and Berkovich geometry is *conjectural*. This section does not claim that the Giry Kleisli category *is* a Berkovich G-site, that iterating the Giry monad *is* iterating a spectrum construction, that a rational-domain localization *is* probabilistic conditioning, that a mixed distributive law *is* a structure sheaf, or that the shadow operators *are* Grothendieck topologies. Each of these is a *research direction* that requires an explicit construction. The purpose of the section is to state the direction precisely and to identify the first concrete bridge.

13.2 What is constructed, and what is conjectural

The following table separates the components that are established from those that are proposed.

Statement	Status	Reason
Giry Kleisli morphisms are Markov kernels	Established	Standard probabilistic interpretation of the Giry monad.
Probability distributions form convex structures, with measurable structure	Established with qualifications	Giry algebras are not all convex spaces; the classification requires care.
Berkovich affinoid geometry has admissible domains and a G-topology	Established	Standard Berkovich analytic geometry.
A rational-domain localization is conditioning on $f \neq 0$	**Incorrect as stated**	Analytic localization is algebraic; probabilistic conditioning is measure-theoretic.
The Giry Kleisli category is a Berkovich G-topological site	**Not established**	A Grothendieck topology on $\mathrm{Kl}(G)$ must be specified and verified.
$G^2(X)$ and $G^3(X)$ correspond to iterated Berkovich spectra	**Unsupported**	Iterating G and iterating a spectrum functor are different operations.
A mixed distributive law is a structure sheaf or sheaf condition	**Incorrect without a new construction**	Beck axioms and sheaf gluing have different types.
Beck defects are obstruction cocycles	Research hypothesis	A cochain complex and cocycle condition must be defined.
L and R are two G-topologies	**Unsupported**	Shadow operators do not automatically define Grothendieck topologies.

The first three rows are established mathematical ingredients. The remaining rows are *proposed*, not proved, and require an explicit construction before they can be tested.

13.3 Why the claimed identifications do not follow

A Berkovich point is not automatically a probabilistic state. A Berkovich point of a k-affinoid algebra A is a bounded multiplicative seminorm on A. A probabilistic state is a positive normalized linear functional on a suitable function space, or a probability measure. The two notions can sometimes be related by a carefully specified construction, but they are not interchangeable by definition.

A rational domain is not a conditioning operation. A rational domain is defined by analytic inequalities

$$\{x \in \mathcal{M}(A) : |f(x)| \leq |g(x)|,\ g(x) \neq 0\},$$

and its associated localization is an algebraic operation. It is not generally obtained by conditioning a probability measure on the event $f \neq 0$. A meaningful bridge would require a probability model on the analytic space, a measurable event corresponding to the rational domain, and a proof that the algebraic localization corresponds to the probabilistic conditioning.

The Kleisli category is not automatically a site. The category $\mathrm{Kl}(G)$ has measurable spaces as objects and Markov kernels as morphisms. This does not, by itself, give it a Grothendieck topology. To make it a site, one must specify covering sieves or covering families and verify the Grothendieck axioms. The proposal's phrase "families of Markov kernels that jointly cover the target" is not yet a definition.

A mixed distributive law is not a structure sheaf. A mixed distributive law $\lambda : GC \Rightarrow CG$ is a natural transformation satisfying the Beck axioms. A sheaf condition says that compatible local sections glue uniquely. The two structures can be related only after a site and a structure sheaf have been constructed and the interaction has been proved.

Shadow operators are not Grothendieck topologies. A shadow operator is a conditional expectation. It does not, by itself, define a Grothendieck topology. To interpret the shadows as topologies, one must specify the covering sieves and verify the Grothendieck axioms.

13.4 A staged programme

The proposal is best understood as a staged programme.

(1) **Construct the probabilistic side rigorously.** Work with standard Borel spaces and the Giry monad. Define the posterior levels $G(B), G^2(B), G^3(B)$, the context maps, and the L/R shadow operators with their domains and codomains.

(2) **Define the local-to-global structure.** Specify what counts as a local posterior model, a restriction map, an admissible cover, and compatible data on overlaps. Prove the covering axioms rather than assuming them.

(3) **Build a genuine analytic bridge.** Choose an appropriate affinoid algebra or analytic space and define a functor or comparison map connecting its analytic domains to the probabilistic models. Prove which probabilistic constructions are preserved.

(4) **Relate the two theories through a theorem.** Only after the bridge exists should one ask whether a Beck diagram corresponds to a gluing diagram, whether its failure defines an obstruction, or whether the L/R structure induces two compatible topologies.

The most productive next step is *not* to compute sheaf cohomology. It is to construct *one explicit bridge* between a well-defined finite or standard-Borel posterior tower and a genuine analytic localization structure. Once that exists, the proposed connection can be tested mathematically rather than asserted by analogy.

13.5 A finite test case

The simplest setting for the bridge is the finite case. Let $B = \{0, 1, \ldots, n-1\}$, so that the Giry monad agrees with the finite distribution monad D. The Kleisli category is the category of finite stochastic matrices. On the analytic side, take the algebra $A = \mathbb{R}^B$ of real-valued functions on B, with the supremum norm. Its Berkovich spectrum $\mathcal{M}(A)$ is a finite set, and its G-topology is generated by the rational domains of A.

Remark 13.1 (The finite bridge as an open problem). A concrete open problem is to define a functor

$$\Phi : \mathrm{Kl}(\mathsf{D}) \to \mathsf{An}_{\mathbb{R}}$$

from the Kleisli category of the finite distribution monad to Berkovich analytic spaces over $\mathbb{R}$, such that:

(i) $\Phi(B) = \mathcal{M}(A)$ for the algebra $A = \mathbb{R}^B$;

(ii) Φ sends a Markov kernel $k : B \to \mathsf{D}(B')$ to an analytic map $\Phi(k) : \mathcal{M}(A) \to \mathcal{M}(A')$;

(iii) Φ preserves the Kleisli composition up to a specified natural transformation;

(iv) on the Kleisli category, Φ induces a Grothendieck topology whose covering sieves correspond to the analytic G-topology.

Whether such a functor exists, and what it preserves, is the first concrete question of the programme. A positive answer would provide the bridge; a negative answer would show that the analogy does not extend beyond the level of a structural metaphor.

The finite case is the right starting point because:

- The Giry monad is the finite distribution monad, which is computationally tractable.
- The Berkovich spectrum of $\mathbb{R}^B$ is a finite set with a combinatorial G-topology.
- The Kleisli category is the category of stochastic matrices, which has a well-understood algebraic structure.
- The proposed functor Φ can be tested by direct computation.

13.6 The standard-Borel case

Once the finite case is settled, the next step is the standard-Borel case. Let $B = [0, 1]$, so that the Giry monad acts on standard Borel spaces. The Kleisli category is the category of Markov kernels on $[0, 1]$. On the analytic side, take the algebra $A = C([0, 1])$ of continuous functions on $[0, 1]$, with the supremum norm. Its Berkovich spectrum $\mathcal{M}(A)$ is a compact Hausdorff space, and its G-topology is generated by the rational domains of A.

The proposed functor Φ would send:

- Objects: a standard Borel space X to the Berkovich spectrum $\mathcal{M}(C(X))$.
- Morphisms: a Markov kernel $k : X \to G(Y)$ to an analytic map $\Phi(k) : \mathcal{M}(C(X)) \to \mathcal{M}(C(Y))$.

Whether such a functor exists, and what it preserves, depends on the interaction between the measure-theoretic and analytic structures. This is the second concrete question of the programme.

13.7 What the proposal does not claim

The proposal does not claim:

(i) that the Giry Kleisli category is a Berkovich G-site;

(ii) that iterating the Giry monad is iterating a spectrum construction;

(iii) that a rational-domain localization is probabilistic conditioning;

(iv) that a mixed distributive law is a structure sheaf;

(v) that the shadow operators are Grothendieck topologies;

(vi) that the Beck defects are obstruction cocycles.

Each of these is a research direction requiring an explicit construction. The purpose of the section is to state the direction precisely and to identify the first concrete bridge.

13.8 A defensible central statement

The following is a defensible central statement for the proposal.

> The Giry monad provides a probabilistic tower of spaces of measures, while Berkovich geometry provides a theory of analytic localization and admissible gluing. We propose investigating whether an appropriate analytic realization of the posterior tower can translate its shadow-compatibility and lifting problems into local-to-global questions. Such a realization requires an explicit site, a suitable analytic algebra, and a comparison construction; no identification of the Giry Kleisli category with a Berkovich G-site is assumed.

This statement preserves the idea without presenting the analogy as a theorem.

13.9 Why the connection is plausible

Despite the gaps, the connection is plausible for three reasons.

Both have a notion of admissible localization. In Berkovich geometry, the rational domains are the admissible localizations. In the Giry Kleisli category, the conditional distributions are the admissible localizations. The two operations are different, but they share the structure of "restrict to a subdomain."

Both have a notion of admissible gluing. In Berkovich geometry, the G-topology specifies which coverings are admissible. In the Giry Kleisli category, the composition of Markov kernels specifies which local data can be glued. The two operations are different, but they share the structure of "compose local data."

Both have a notion of obstruction. In Berkovich geometry, the cohomology of the G-topology obstructs global sections. In the Giry tower, the Beck defects obstruct the mixed distributive law. The two obstructions are different, but they share the structure of "failure of gluing."

The proposal is to investigate whether these three shared structures can be made precise by a single functor. This is the research direction, and it is the correct level of claim for the current state of the programme.

13.10 Relation to the double-category paper

The double-category paper's two shadows L and R are the candidate G-topologies of the proposed framework. If the framework is realized, the interchange law for the two shadows is the compatibility of the two G-topologies, and the (L, R)-compatible subcategory is the subcategory on which the two topologies agree.

This is a *proposed* interpretation, not a theorem. It is consistent with the double-category paper's construction, and it suggests a direction for future work.

14 The Finite Stochastic-to-Convex Bridge

The double-category construction of Section 11 uses the Kleisli category of the finite distribution monad as its horizontal layer. This section constructs a covariant functor from that Kleisli category to a concrete category of finite-dimensional probability simplices and affine maps. The functor is *provable* with the current machinery, and it makes the tower's diagrammatic structure concrete and computable. It does *not* land in Berkovich analytic spaces; the analytic extension remains an open problem.

14.1 Setup

Definition 14.1 (Finite stochastic kernels)**.** Let **FinStoch** be the category whose:

- **objects** are finite sets $X, Y, Z, \ldots$;
- **morphisms** $K : X \to Y$ are stochastic kernels
$$K(y \mid x) \geq 0, \qquad \sum_{y \in Y} K(y \mid x) = 1;$$
- **composition** is matrix multiplication:
$$(L \circ K)(z \mid x) = \sum_{y \in Y} L(z \mid y)\, K(y \mid x);$$
- **identities** are the Kronecker deltas
$$I_X(x' \mid x) = \begin{cases} 1, & x' = x, \\ 0, & x' \neq x. \end{cases}$$

The category **FinStoch** is the Kleisli category of the finite distribution monad D restricted to finite sets.

Definition 14.2 (Probability simplices)**.** For a finite set X, define the *probability simplex*
$$\Delta_X = \Big\{ p \in \mathbb{R}^X : p(x) \geq 0,\ \sum_{x \in X} p(x) = 1 \Big\}.$$

Let **Conv** be the category whose:

- **objects** are the simplices Δ_X for finite sets X;
- **morphisms** are affine maps $T : \Delta_X \to \Delta_Y$, i.e. maps satisfying $T(tp + (1-t)q) = tT(p) + (1-t)T(q)$ for all $p, q \in \Delta_X$ and $t \in [0, 1]$.

Every stochastic kernel $K : X \to Y$ induces a map
$$T_K : \Delta_X \longrightarrow \Delta_Y, \qquad T_K(p)(y) = \sum_{x \in X} p(x)\, K(y \mid x). \tag{3}$$

14.2 The theorem

Theorem 14.3 (Finite stochastic-to-convex bridge)**.** *The assignment*

$$T : \mathbf{FinStoch} \longrightarrow \mathbf{Conv}, \qquad X \longmapsto \Delta_X, \qquad K \longmapsto T_K,$$

is a covariant functor. It preserves identities and Kleisli composition, and each induced map T_K is affine.

The proof is given by the four lemmas below.

Lemma 14.4 (Well-definedness)**.** *For every $K : X \to Y$ and every $p \in \Delta_X$, $T_K(p) \in \Delta_Y$.*

Proof. Nonnegativity: for each $y \in Y$,

$$T_K(p)(y) = \sum_{x \in X} p(x)K(y \mid x) \geq 0,$$

because every term is nonnegative.

Normalization: since K is stochastic,

$$\sum_{y \in Y} T_K(p)(y) = \sum_{y \in Y} \sum_{x \in X} p(x)K(y \mid x) = \sum_{x \in X} p(x) \sum_{y \in Y} K(y \mid x) = \sum_{x \in X} p(x) = 1.$$

Hence $T_K(p) \in \Delta_Y$. □

Lemma 14.5 (Affineness)**.** *For every $K : X \to Y$, every $p, q \in \Delta_X$, and every $t \in [0, 1]$,*

$$T_K(tp + (1-t)q) = tT_K(p) + (1-t)T_K(q).$$

Proof. For each $y \in Y$,

$$\begin{aligned} T_K(tp + (1-t)q)(y) &= \sum_{x \in X} \big(tp(x) + (1-t)q(x)\big)K(y \mid x) \\ &= t \sum_{x \in X} p(x)K(y \mid x) + (1-t) \sum_{x \in X} q(x)K(y \mid x) \\ &= tT_K(p)(y) + (1-t)T_K(q)(y). \end{aligned}$$

Since this holds for every y, the two vectors are equal. □

Lemma 14.6 (Identity preservation)**.** *For every finite set X, $T_{I_X} = \mathrm{id}_{\Delta_X}$.*

Proof. For $p \in \Delta_X$ and $x' \in X$,

$$T_{I_X}(p)(x') = \sum_{x \in X} p(x)I_X(x' \mid x) = p(x').$$

Hence $T_{I_X}(p) = p$ for every p. □

Lemma 14.7 (Composition preservation)**.** *For composable stochastic kernels $K : X \to Y$ and $L : Y \to Z$,*

$$T_{L \circ K} = T_L \circ T_K.$$

Proof. For $p \in \Delta_X$ and $z \in Z$,

$$\begin{aligned} T_{L\circ K}(p)(z) &= \sum_{x\in X} p(x)(L\circ K)(z \mid x) \\ &= \sum_{x\in X} p(x) \sum_{y\in Y} L(z \mid y)K(y \mid x) \\ &= \sum_{y\in Y} L(z \mid y) \sum_{x\in X} p(x)K(y \mid x) \\ &= \sum_{y\in Y} T_K(p)(y)L(z \mid y) \\ &= T_L(T_K(p))(z). \end{aligned}$$

Since this holds for every z and every p, the two maps are equal. □

Proof of Theorem 14.3. Lemmas 14.4 and 14.5 show that each stochastic kernel induces a well-defined affine map. Lemma 14.6 establishes preservation of identities, and Lemma 14.7 establishes preservation of composition. Hence T is a covariant functor. □

14.3 Continuity and non-expansiveness

Equip each simplex with the topology inherited from its ambient Euclidean space.

Proposition 14.8 (Continuity and ℓ^1-non-expansiveness)**.** *For every $K : X \to Y$, the induced map T_K is continuous. In fact, for all $p, q \in \Delta_X$,*

$$\|T_K(p) - T_K(q)\|_1 \le \|p - q\|_1.$$

Proof. Because T_K is given by finite matrix multiplication, it is continuous. For the non-expansiveness,

$$\begin{aligned} \|T_K(p) - T_K(q)\|_1 &= \sum_{y\in Y} \Big| \sum_{x\in X} (p(x) - q(x))K(y \mid x) \Big| \\ &\le \sum_{y\in Y} \sum_{x\in X} |p(x) - q(x)|K(y \mid x) \\ &= \sum_{x\in X} |p(x) - q(x)| \sum_{y\in Y} K(y \mid x) \\ &= \|p - q\|_1. \end{aligned}$$

Hence T_K is non-expansive. □

14.4 What the bridge establishes

Claim	Status
Stochastic kernels induce affine maps of probability simplices	Proven
Identities and Kleisli composition are preserved	Proven
Induced maps are continuous and ℓ^1-non-expansive	Proven
A functor exists into finite-dimensional convex spaces	Proven
A functor exists into Berkovich analytic spaces	**Not proven**
Induced maps are multiplicative algebra homomorphisms	**Not required, generally false**

The bridge is *concrete*: finite sets, simplices, affine maps. It is *explicit*: the construction is given by equation (3). It is *provable*: the four lemmas are complete.

14.5 What the bridge does not establish

The bridge does *not* establish:

(i) A functor into Berkovich analytic spaces.

(ii) That the induced maps are multiplicative.

(iii) That the simplices carry a Berkovich G-topology compatible with the Kleisli composition.

The last two rows of the table are the crucial disclaimers:

- **Not Berkovich.** The functor lands in **Conv**, not in the category of Berkovich analytic spaces. To land in Berkovich, one would need to specify a target category of analytic spaces, construct an analytic object for each simplex, construct an analytic morphism for each affine map, and prove that the construction preserves identities and composition. This is a separate, substantial construction.
- **Not multiplicative.** The induced map T_K is linear (affine), not multiplicative. It preserves positivity and the constant function 1, but it does not preserve products. Excluding multiplication is the correct choice: it removes the obstruction that a stochastic expectation operator is generally not multiplicative.

14.6 Relation to the tower

The tower's base-relative construction works over the finite distribution monad D. The Kleisli category of D restricted to finite sets is exactly **FinStoch**. The functor T sends:

- a finite set B to the simplex Δ_B;
- a Markov kernel $K : B \to \mathsf{D}(B')$ to the affine map $T_K : \Delta_B \to \Delta_{B'}$;
- the tower's levels $\mathsf{D}(B), \mathsf{D}^2(B), \mathsf{D}^3(B), \ldots$ to the iterated simplices $\Delta_B, \Delta_{\Delta_B}, \Delta_{\Delta_{\Delta_B}}, \ldots$

Under T, the tower's two shadows L and R become affine endomorphisms of Δ_{Δ_B}, and the interchange law for L and R becomes the commutativity of the corresponding affine maps. The functor T preserves the tower's diagrammatic structure: if a Beck diagram commutes in **FinStoch**, it commutes in **Conv**, and vice versa. This is because T is a functor.

14.7 Relation to the double category

The double-category construction of Section 11 can be transported along T:

Double-category construction	Image under T
Objects $c \in \mathsf{Coh}(B)$	Simplices Δ_B
Horizontal 1-cells $f : c \to c'$	Affine maps $T_f : \Delta_B \to \Delta_{B'}$
Vertical 1-cells L, R	Affine endomorphisms of Δ_{Δ_B}
2-cells	Comparisons of affine maps
Interchange law	Commutativity of affine maps
Lifting functor F	The functor T itself

So T transports the double-category paper's structure from **FinStoch** to **Conv**. The double-category paper's constructions are *provable* in the target category **Conv**. This makes the double-category paper's constructions concrete and computable.

14.8 The analytic extension as an open problem

Conjecture 14.9 (Analytic extension). There exists a functor

$$\Phi : \mathbf{FinStoch} \longrightarrow \mathsf{An}_k$$

from the Kleisli category of the finite distribution monad to Berkovich analytic spaces over a non-Archimedean field k, such that:

(i) $\Phi(B) = \mathcal{M}(A_B)$ for an affinoid algebra A_B associated with B;

(ii) $\Phi(K) : \mathcal{M}(A_B) \to \mathcal{M}(A_{B'})$ is an analytic morphism lifting T_K;

(iii) Φ preserves Kleisli composition up to a specified natural transformation;

(iv) on **FinStoch**, Φ induces a Grothendieck topology whose covering sieves correspond to the analytic G-topology.

Whether such a functor exists, and what it preserves, is the first concrete question of the analytic programme.

The functor T of Theorem 14.3 is the *first stage* of the bridge. The analytic extension is a *separate, substantial construction* and remains an open problem.

14.9 The standard-Borel case

The finite case is the right starting point because:

- the Giry monad agrees with the finite distribution monad on finite sets;
- the Kleisli category is the category of finite stochastic matrices, which is computationally tractable;

- the simplices Δ_B are finite-dimensional, so the functor T can be tested by direct computation.

The standard-Borel case is the natural next step. Let $B = [0, 1]$, so that the Giry monad acts on standard Borel spaces. The Kleisli category is the category of Markov kernels on $[0, 1]$. The functor T would send a Markov kernel $K : X \to G(Y)$ to an affine map $T_K : \Delta_X \to \Delta_Y$ between the spaces of probability measures on X and Y. Whether this extension preserves the relevant structure (e.g. disintegration, conditional independence) is a further open problem.

14.10 Conclusion

The finite stochastic-to-convex bridge is a *working theorem* that is valid as written. It gives:

- a covariant functor $T : \mathbf{FinStoch} \to \mathbf{Conv}$;
- preservation of identities and Kleisli composition;
- continuity and ℓ^1-non-expansiveness of the induced maps;
- a concrete, computable home for the tower's constructions.

It does *not* land in Berkovich analytic spaces, and it does *not* require multiplicative maps. The analytic extension is a separate, substantial construction, stated as Conjecture 14.9. Until that construction is supplied, the bridge is the complete working proof of the finite stochastic-to-convex connection.

15 A Wasserstein–Berkovich Bridge for the Finite Stochastic Tower

This section constructs a covariant functor from the Kleisli category of the finite distribution monad to the category of Berkovich analytic spaces over a non-Archimedean field, and equips it with a Wasserstein–Gel'fand compatibility condition. The construction is a refinement of the finite stochastic-to-convex bridge of Section **??**: the convex target is replaced by a Berkovich target, and the ℓ^1 topology on simplices is replaced by the W_1 Wasserstein metric on measures. The functor itself is proved (Theorem 15.4). The topological compatibility and the cohomological interpretation of the Beck defects are stated as obligations (Obligations 15.6, 15.6) and remain open.

15.1 Motivation

The finite stochastic-to-convex bridge of Section **??** sends a finite set X to its probability simplex Δ_X, and a stochastic kernel $K : X \to Y$ to the affine map $T_K : \Delta_X \to \Delta_Y$. The target category **Conv** has finite-dimensional convex objects and continuous affine morphisms. It is concrete, computable, and provable.

The bridge does not reach Berkovich analytic spaces. The reason is structural: a Berkovich space is a space of bounded multiplicative seminorms on a Banach algebra, not a space of probability measures. The two constructions agree only in degenerate cases, and the simplex is not the Berkovich spectrum of any natural algebra associated with a finite set.

This section proposes a different route. Rather than trying to identify the simplex with a Berkovich space, we construct a functor that *refines* the finite bridge by passing through a Berkovich analytification of the simplex. The refinement is not an identification. The Berkovich space is strictly larger than the simplex; the simplex sits inside it as the classical locus. The point of the construction is that the Berkovich space carries a G-topology, and the G-topology is the structure that supports the sheaf-theoretic and cohomological machinery that the Beck defects require.

The metric question is settled as follows. On the probability-measure side, we use the W_1 Wasserstein metric, which metrizes the weak topology on Δ_X for finite X. On the Berkovich side, we use the Gel'fand topology as the underlying topological structure and the G-topology as the sheaf-theoretic structure. The bridge is *not* metric-to-metric. It is metric-on-measures to topology-plus-G-topology on seminorms. The Wasserstein metric controls the classical locus; the Gel'fand topology controls the ambient Berkovich space; the G-topology controls admissible localization and gluing.

15.2 The source category

Let **FinStoch** be the Kleisli category of the finite distribution monad $\mathbb{D}$ on finite sets. Objects are finite sets $X, Y, Z, \ldots$. A morphism $K : X \to Y$ is a stochastic kernel

$$K(y \mid x) \geq 0, \qquad \sum_{y \in Y} K(y \mid x) = 1.$$

Composition is matrix multiplication:

$$(L \circ K)(z \mid x) = \sum_{y \in Y} L(z \mid y) K(y \mid x).$$

Identities are Kronecker deltas. This is the same source category as in Section **??**.

15.3 The target category

Let k be a non-Archimedean field, complete with respect to a nontrivial absolute value $|\cdot|_k$. Let $\mathbf{An}_k$ denote the category of Berkovich k-analytic spaces. Objects are locally ringed spaces over k that are locally isomorphic to affinoid domains, equipped with a G-topology and a G-sheaf of k-affinoid algebras. Morphisms are morphisms of locally ringed G-topological spaces over k.

Each object $X \in \mathbf{An}_k$ carries two topological structures:

- The **Gel'fand topology** τ_{Gel}, the weak topology generated by evaluation maps $x \mapsto |f|_x$ for $f \in \mathcal{O}_X(X)$.
- The **G-topology** $\mathcal{T}_X$, the Grothendieck topology of admissible open affinoid domains.

For an affinoid k-algebra $\mathcal{A}$, write $\mathcal{M}(\mathcal{A})$ for its Berkovich spectrum. A point $x \in \mathcal{M}(\mathcal{A})$ is a bounded multiplicative seminorm $|\cdot|_x : \mathcal{A} \to \mathbb{R}_{\geq 0}$. The Berkovich spectrum functor $\mathcal{M} : \mathbf{Aff}_k^{\mathrm{op}} \to \mathbf{An}_k$ sends a bounded k-algebra homomorphism $\varphi : \mathcal{A} \to \mathcal{B}$ to the morphism $\mathcal{M}(\varphi) : \mathcal{M}(\mathcal{B}) \to \mathcal{M}(\mathcal{A})$ defined by $|\bar{a}|_{\mathcal{M}(\varphi)(y)} = |\varphi(a)|_y$.

15.4 The simplex algebra and its Berkovich analytification

Fix a finite set X. Let Δ_X be the probability simplex

$$\Delta_X = \left\{ p \in \mathbb{R}^X : p(x) \geq 0, \sum_{x \in X} p(x) = 1 \right\},$$

equipped with the Euclidean topology, which coincides with the weak topology on Δ_X. Let

$$\mathcal{B}_X = C(\Delta_X)$$

be the real Banach algebra of continuous real-valued functions on Δ_X, with the supremum norm. This is a commutative unital real Banach algebra.

Define the k-affinoid algebra

$$\mathcal{A}_X = \mathcal{B}_X \otimes_{\mathbb{R}} k,$$

with the tensor-product norm. This is a commutative unital k-Banach algebra. Its Berkovich spectrum $\mathcal{M}(\mathcal{A}_X)$ is the **Berkovich analytification of the simplex** over k.

Remark 15.1 (Relation to the simplex). When $k = \mathbb{R}$, the Berkovich spectrum $\mathcal{M}(\mathcal{A}_X)$ is the Gel'fand spectrum of $\mathcal{B}_X$, which is Δ_X itself. When k is non-Archimedean, $\mathcal{M}(\mathcal{A}_X)$ is strictly larger than Δ_X: it contains the classical locus Δ_X as a dense subset, but also contains non-Archimedean points. The classical locus is the image of the inclusion

$$\Psi_X : \Delta_X \hookrightarrow \mathcal{M}(\mathcal{A}_X).$$

The Gel'fand topology on $\mathcal{M}(\mathcal{A}_X)$ restricts to the Euclidean topology on Δ_X.

Remark 15.2 (Why not the simplex algebra). One might instead take $\mathcal{A}_X = k\langle t_x : x \in X\rangle/(\sum_x t_x - 1)$, the k-affinoid algebra of the closed simplex. Its Berkovich spectrum is the Berkovich simplex, which also contains Δ_X as the classical locus. The two choices differ in their non-Archimedean points but agree on the classical locus. We use $C(\Delta_X) \otimes_{\mathbb{R}} k$ because it is the algebra whose Berkovich spectrum is the *analytification* of the simplex, which is the standard construction in Berkovich's theory for analytifying algebraic varieties over non-Archimedean fields. The simplex algebra is the coordinate ring of the simplex as an algebraic variety; the analytification of that variety is the Berkovich spectrum of the tensor product of the coordinate ring with k, not of $C(\Delta_X) \otimes_{\mathbb{R}} k$. Both are legitimate, and both yield functors. We record the choice because it affects the G-topology: the analytification of the algebraic simplex has a G-topology generated by rational domains in the algebraic coordinates, which is the G-topology that the Beck defects will use.

15.5 The functor on objects and morphisms

Definition 15.3 (Wasserstein–Berkovich bridge)**.** Define

$$\Phi : \mathbf{FinStoch} \to \mathbf{An}_k$$

on objects by

$$\Phi(X) = \mathcal{M}(\mathcal{A}_X) = \mathcal{M}(C(\Delta_X) \otimes_{\mathbb{R}} k),$$

the Berkovich analytification of the simplex.

On morphisms, let $K : X \to Y$ be a stochastic kernel. Define a bounded k-algebra homomorphism

$$K^* : \mathcal{A}_Y \to \mathcal{A}_X$$

by extending $K^*(\mathrm{ev}_y) = \sum_{x \in X} K(y \mid x)\, \mathrm{ev}_x$ from the coordinate functions $\mathrm{ev}_y \in C(\Delta_Y)$ to the tensor product. Here $\mathrm{ev}_y(p) = p(y)$ is the y-th coordinate function on the simplex. The right-hand side is an element of $C(\Delta_X)$, and the assignment preserves the unit because $\sum_y \sum_x K(y \mid x)\mathrm{ev}_x = \sum_x \mathrm{ev}_x = 1$. Define

$$\Phi(K) = \mathcal{M}(K^*) : \mathcal{M}(\mathcal{A}_X) \to \mathcal{M}(\mathcal{A}_Y),$$

where $\mathcal{M}$ is the Berkovich spectrum functor.

Theorem 15.4 (Functoriality of Φ)**.** *The assignment $\Phi : \mathbf{FinStoch} \to \mathbf{An}_k$ is a covariant functor. It preserves identities and Kleisli composition.*

Proof. The assignment $X \mapsto \mathcal{A}_X$ is functorial on objects. The assignment $K \mapsto K^*$ is contravariantly functorial:

$$(L \circ K)^*(\mathrm{ev}_z) = \sum_x (L \circ K)(z \mid x)\mathrm{ev}_x = \sum_x \sum_y L(z \mid y)K(y \mid x)\mathrm{ev}_x = \sum_y L(z \mid y)K^*(\mathrm{ev}_y) = K^*(L^*(\mathrm{ev}_z)).$$

Since the coordinate functions generate $C(\Delta_Z)$ as a Banach algebra, the identity extends to all of $\mathcal{A}_Z$ by continuity. Identities: $(\mathrm{id}_X)^*(\mathrm{ev}_y) = \sum_x \delta_{xy}\mathrm{ev}_x = \mathrm{ev}_y$, so $(\mathrm{id}_X)^* = \mathrm{id}_{\mathcal{A}_X}$. The Berkovich spectrum functor $\mathcal{M}$ is contravariant on $\mathbf{Aff}_k$, so the composite $\mathcal{M} \circ (-)^*$ is covariant. □

Remark 15.5 (Comparison with the finite bridge). The functor $T : \mathbf{FinStoch} \to \mathbf{Conv}$ of Section **??** sends X to Δ_X and K to T_K. The functor Φ sends X to the Berkovich analytification of Δ_X and K to the Berkovich morphism induced by K^*. On the classical locus, $\Phi(K)$ restricts to T_K: the diagram

$$\begin{array}{ccc} \Delta_X & \xrightarrow{\Psi_X} & \mathcal{M}(\mathcal{A}_X) \\ \downarrow{\scriptstyle T_K} & & \downarrow{\scriptstyle \Phi(K)} \\ \Delta_Y & \xrightarrow{\Psi_Y} & \mathcal{M}(\mathcal{A}_Y) \end{array}$$

commutes, because Ψ_X and Ψ_Y are the inclusions of the classical loci and $\Phi(K)$ restricts to the map induced by K^* on characters, which is T_K. The functor Φ is therefore a refinement of T, not a replacement.

15.6 Wasserstein–Gel'fand compatibility

We now equip the probability-measure side with the W_1 Wasserstein metric and state the compatibility condition.

Definition 15.6 (Wasserstein metric on the simplex)**.** For $p, q \in \Delta_X$, define

$$W_1(p, q) = \sup_{\|f\|_{\mathrm{Lip}} \le 1} \left(\sum_{x \in X} f(x)p(x) - \sum_{x \in X} f(x)q(x) \right),$$

where the supremum is over real-valued 1-Lipschitz functions on X with respect to the discrete metric. For finite X, W_1 metrizes the weak topology on Δ_X.

Proposition 15.7 (Wasserstein–Gel'fand compatibility on the classical locus)**.** *For every stochastic kernel $K : X \to Y$, the restriction of $\Phi(K)$ to the classical locus is continuous and W_1-Lipschitz. In particular, if $p_n \to p$ in W_1 on Δ_X, then $\Phi(K)(\Psi_X(p_n)) \to \Phi(K)(\Psi_X(p))$ in the Gel'fand topology on $\mathcal{M}(\mathcal{A}_Y)$.*

Proof. The restriction of $\Phi(K)$ to the classical locus is T_K, by Remark 15.5. The map T_K is continuous and ℓ^1-nonexpansive by Proposition 14.8 of Section **??**. On finite simplices, the W_1 metric and the ℓ^1 metric are equivalent up to a constant depending on $|X|$. Hence T_K is W_1-Lipschitz. The inclusion Ψ_Y is a topological embedding, so Gel'fand-convergence of $\Psi_Y(T_K(p_n))$ follows from W_1-convergence of $T_K(p_n)$. □

[G-topology pullback along $\Phi(K)$] Prove that the G-topology $\mathcal{T}_{\Phi(Y)}$ on $\Phi(Y)$ pulls back along $\Phi(K)$ to a G-topology $\mathcal{T}_{\Phi(K)^*}$ on $\Phi(X)$:

$$\mathcal{T}_{\Phi(K)^*} = \{\Phi(K)^{-1}(U) : U \in \mathcal{T}_{\Phi(Y)}\},$$

and that this pulled-back G-topology is compatible with the Wasserstein topology on the classical locus, in the following sense:

(i) **Admissibility.** The preimage of an admissible open affinoid domain under $\Phi(K)$ is admissible open in $\Phi(X)$.

(ii) **Gluing.** If $\{U_i\}$ is an admissible cover of $\Phi(Y)$, then $\{\Phi(K)^{-1}(U_i)\}$ is an admissible cover of $\Phi(X)$ in the pulled-back G-topology.

(iii) **Metric compatibility.** On the classical locus, the W_1-topology is compatible with the pulled-back G-topology: if $p_n \to p$ in W_1, then for every admissible open $U \subseteq \Phi(Y)$ containing $\Phi(K)(\Psi_X(p))$, there exists N such that $\Psi_X(p_n) \in \Phi(K)^{-1}(U)$ for all $n \geq N$.

Remark 15.8 (Status of Obligation 15.6). Admissibility (i) is automatic: $\Phi(K)$ is a morphism of $\mathbf{An}_k$, and morphisms of Berkovich spaces pull back admissible opens to admissible opens by definition of the G-topology. Gluing (ii) follows from admissibility and the functoriality of the G-topology. Metric compatibility (iii) is the nontrivial obligation. It requires that the W_1-topology on Δ_X is finer than the topology induced by the Gel'fand topology on $\Phi(X)$ restricted to the classical locus, and that the G-topology's admissible opens are open in the Gel'fand topology (which they are, by definition of the G-topology). The obligation is therefore a statement about the interaction of the Wasserstein topology with the Gel'fand topology, not about the G-topology per se.

[Beck defects as G-cohomology classes] Construct the Čech complex of the G-topology $\mathcal{T}_{\Phi(Y)}$, and define a map from Beck defects of a candidate shift map $\lambda : DC \Rightarrow CD$ to Čech 1-cocycles on $\Phi(Y)$. Prove that the cocycle is a coboundary if and only if the Beck defect admits a factorization through the shadows L and R.

Remark 15.9 (Status of Obligation 15.6). This is the deepest obligation and the one that connects the bridge to the Beck-defect program of Sections 6 and 13. It requires the construction of the Čech complex, a map from defects to cocycles, and a proof that the vanishing of the cocycle is equivalent to the factorization. None of these is currently established. The purpose of the Berkovich bridge is to give the obstruction theory a precise home; the construction of the theory itself remains open.

15.7 What the bridge establishes and what it does not

The following table separates the components that are proved from those that are proposed.

Statement	**Status**
Functor $\Phi : \mathbf{FinStoch} \to \mathbf{An}_k$ exists	Proved (Theorem 15.4)
Φ refines the finite bridge T on the classical locus	Proved (Remark 15.5)
W_1-Gel'fand compatibility on the classical locus	Proved (Proposition 15.7)
Admissibility of G-topology pullback	Proved (Remark 15.9 (i))
Gluing of pulled-back G-topology	Proved (Remark 15.9 (ii))
Metric compatibility of pulled-back G-topology	Open (Obligation 15.6 (iii))
Beck defects as Čech cohomology classes	Open (Obligation 15.6)
Resolution of the inheritance obligation	Not addressed by this section
Full double-category construction	Not addressed by this section

What the bridge establishes. It constructs a genuine functor from the Kleisli category of the finite distribution monad to the category of Berkovich analytic spaces over a non-Archimedean field. The functor is a refinement of the finite stochastic-to-convex bridge of Section **??**: on the classical locus, it agrees with the finite bridge, and the Wasserstein metric on the simplex is compatible with the Gel'fand topology on the Berkovich space. The bridge equips the tower's finite layer with a

G-topology, which is the structure required for the sheaf-theoretic and cohomological interpretation of the Beck defects.

What the bridge does not establish. It does not construct the Čech complex of the G-topology, and it does not prove that the Beck defects are cohomology classes. It does not resolve the inheritance obligation of Remark 4.9: the commutation of the measure-valued shadows L and R is still a separate obligation. It does not construct the full double category of Section 11. It does not extend to the standard-Borel or continuous Giry case; the functor is finite.

No metric on the Berkovich side. The bridge is not metric-to-metric. The Wasserstein metric is defined on the probability simplex; the Berkovich space carries the Gel'fand topology and the G-topology, neither of which is a metric. The hyperbolic metric on the Berkovich tree is not used, because it is not defined on the simplex and does not metrize the weak topology on measures. The correct structure on the Berkovich side is topology plus G-topology, not metric.

15.8 Concrete next steps

The following are the smallest defensible steps toward realizing the bridge.

1. **Prove Obligation 15.6 (iii) for finite X, Y.** This is a finite computation. The G-topology on $\mathcal{M}(\mathcal{A}_X)$ for finite X is combinatorial: admissible opens are finite unions of rational domains defined by inequalities $|f| \leq |g|$ for $f, g \in \mathcal{A}_X$. The pullback along $\Phi(K)$ can be computed explicitly, and the compatibility with the W_1-topology can be checked on the finite classical locus.

2. **Construct the Čech complex of the G-topology** for the finite test case $B = \{0, \ldots, n-1\}$, $A = \mathbb{R}^B$. In this case the G-topology is combinatorial and finite, and the Čech complex can be written down explicitly. The Beck defect of a candidate shift map can be computed as a 1-cocycle, and the factorization through L and R can be tested by checking whether the cocycle is a coboundary.

3. **Extend to the standard-Borel case.** Replace X with $[0, 1]$ and $\mathcal{A}_X$ with $C([0, 1]) \otimes_{\mathbb{R}} k$. The Berkovich spectrum is the analytification of the interval, and the G-topology is generated by rational domains in the coordinate function. The Wasserstein metric on $\mathcal{P}([0, 1])$ metrizes the weak topology, and the compatibility with the Gel'fand topology becomes a statement about the interaction of weak convergence with the analytic topology.

4. **Connect to the double category of Section 11.** The functor Φ transports the horizontal layer of the double category from **FinStoch** to $\mathbf{An}_k$. The vertical layer, consisting of the shadows L and R, must be transported as well. This requires defining the shadows on the Berkovich side, which is a separate construction: conditional expectation on a Berkovich space is not the same as conditional expectation on a simplex, and the two must be related by an explicit comparison map.

5. **State the analytic extension as a conjecture.** The functor Φ is finite. The analytic extension to standard Borel spaces and the Giry monad is a separate conjecture. The finite bridge is the first stage; the analytic extension is the second.

15.9 Conclusion

The Wasserstein–Berkovich bridge is a functor

$$\Phi : \mathbf{FinStoch} \to \mathbf{An}_k$$

that refines the finite stochastic-to-convex bridge of Section **??** by passing through the Berkovich analytification of the probability simplex. The functor is proved (Theorem 15.4). It agrees with the finite bridge on the classical locus (Remark 15.5). The W_1 Wasserstein metric on the simplex is compatible with the Gel'fand topology on the Berkovich space (Proposition 15.7). The G-topology on the Berkovich space pulls back along $\Phi(K)$ to a G-topology on $\Phi(X)$, and admissibility and gluing are automatic (Remark 15.9 (i), (ii)). The metric compatibility of the pulled-back G-topology and the interpretation of the Beck defects as Čech cohomology classes are stated as obligations (Obligations 15.6 (iii), 15.6) and remain open.

The bridge does not provide a metric on the Berkovich side. It provides a topology (Gel'fand), a Grothendieck topology (G-topology), and a functor that connects them to the Wasserstein metric on measures. This is the correct level of structure for the Beck-defect program. The next steps are finite, explicit, and checkable in the finite case, and open in the analytic case.

16 Proposed Work: Wall-Crossings and Secant Varieties of the Gaussian Mixture Tower

This section proposes a programme connecting the fibre stratification of the Gaussian mixture tower to algebraic geometry via secant varieties of Gaussian moment varieties, and lifting the resulting wall-crossing formula to the Berkovich bridge of Section 15. The proposal is stated as a research programme with explicit obligations; no wall-crossing theorem is claimed as established.

16.1 Motivation

The fibre document identifies the discriminant locus

$$\Delta = \bigcup_{i<j} \{\delta_{ij} = 0\}, \qquad \delta_{ij}(x) = \|\mu_i - \mu_j\|^2 + \|\Sigma_i - \Sigma_j\|_F^2,$$

as the set of parameter points in X_n where two Gaussian components coincide. The generic stratum $X_n^{\mathrm{reg}} = X_n \setminus \Delta$ has finite fibres under the forgetful map $\beta_n : X_n \to \mathcal{P}(\mathbb{R}^d)$; the degenerate strata have fibres that can be positive-dimensional.

This is a *wall-crossing* structure: the fibre $F_n(\nu) = \beta_n^{-1}(\nu)$ changes as ν crosses the image of the discriminant locus in the moduli space $\mathcal{M}_n = \beta_n(X_n)$. The formal structure is the same as wall-crossing in Donaldson theory, where the difference of invariants across a wall is a sum over effective classes satisfying a linear condition [**?**], and in Bridgeland stability, where crossing a wall changes the moduli space of stable objects [**?**].

The proposal is to make this structural correspondence precise using *algebraic statistics*: the Gaussian mixture model is an algebraic variety, its parameterization is a moment map, and its image is a secant variety of the Gaussian moment variety [**?**, **?**].

16.2 The algebraic statistics framework

16.2.1 Statistical models as algebraic varieties

A parametric statistical model with parameter space $\Theta \subseteq \mathbb{R}^m$ and sufficient statistic map $s : \mathcal{X} \to \mathbb{R}^M$ is *algebraic* if the parameterization map

$$\theta \mapsto \mathbb{E}_\theta[s(X)]$$

is polynomial or rational. The image of this map is an algebraic variety, called the *model variety*, sitting inside the ambient space $\mathbb{R}^M$ of sufficient statistics [**?**].

For a Gaussian distribution $\mathcal{N}(\mu, \Sigma)$ on $\mathbb{R}^d$, the sufficient statistics up to order two are

$$s(x) = (x, xx^\top) \in \mathbb{R}^d \times \mathrm{Sym}_d,$$

and the moment map is

$$(\mu, \Sigma) \mapsto (\mu, \Sigma + \mu\mu^\top).$$

The image is the *Gaussian moment variety* $\mathcal{G}_{d,2} \subseteq \mathbb{R}^d \times \mathrm{Sym}_d$.

16.2.2 Gaussian mixtures as secant varieties

A mixture of k Gaussians with weights $p_1, \ldots, p_k$ has moment

$$\sum_{i=1}^{k} p_i(\mu_i, \Sigma_i + \mu_i\mu_i^\top).$$

The image of the mixture parameter space $\Theta_{d,k}$ under the moment map is the k-th *secant variety* of the Gaussian moment variety:

$$\mathrm{Sec}_k(\mathcal{G}_{d,2}) = \overline{\left\{ \sum_{i=1}^{k} p_i g_i : g_i \in \mathcal{G}_{d,2},\, p_i \geq 0,\, \sum_i p_i = 1 \right\}}.$$

This is the standard algebraic statistics description of Gaussian mixtures [**?**].

16.2.3 Identifiability and defectivity

The parameter space $\Theta_{d,k}$ has dimension

$$\dim \Theta_{d,k} = kd + k\frac{d(d+1)}{2} + k - 1 = k\left(d + \frac{d(d+1)}{2}\right) + k - 1.$$

The expected dimension of the secant variety is

$$\dim_{\exp} \mathrm{Sec}_k(\mathcal{G}_{d,2}) = \min\left\{ k \cdot \dim \mathcal{G}_{d,2} + k - 1,\, \dim \mathbb{R}^d \times \mathrm{Sym}_d \right\}.$$

The mixture model is *identifiable* when the dimension of the secant variety equals the dimension of the parameter space. It is *defective* when the secant variety is smaller than expected [**?**].

16.3 The tower as a nested sequence of secant varieties

The Gaussian mixture tower X_n is a recursive product of simplices and Gaussian parameter spaces. Its image under the flattening map β_n is a nested sequence of secant varieties:

$$\mathrm{Sec}_1(\mathcal{G}_{d,2}) \subseteq \mathrm{Sec}_4(\mathcal{G}_{d,2}) \subseteq \mathrm{Sec}_{16}(\mathcal{G}_{d,2}) \subseteq \cdots \subseteq \mathrm{Sec}_{4^n}(\mathcal{G}_{d,2}).$$

At each level, the map $\beta_n : X_n \to \mathrm{Sec}_{4^n}(\mathcal{G}_{d,2})$ is the moment map of the level-n hierarchical mixture.

The discriminant locus $\Delta \subseteq X_n$ is the locus where the moment map fails to be injective on the minimal representation. In the algebraic statistics language:

- The **generic stratum** X_n^{reg} is where the secant variety $\mathrm{Sec}_{4^n}(\mathcal{G}_{d,2})$ has the expected dimension and the moment map is generically injective.

- The **degenerate strata** are where the secant variety becomes *defective*: its dimension drops below the expected value, and the fibres of the moment map become positive-dimensional.

The wall-crossing structure of the tower is therefore the *defectivity stratification* of the secant varieties.

16.4 The wall-crossing formula

16.4.1 Formal template from Donaldson theory

In Donaldson theory, the wall-crossing formula for the difference of invariants across a wall is

$$\Phi_{H_+}(\hat{y}) - \Phi_{H_-}(\hat{y}) = \sum_{(a_0,a_1)\in S_0(y,\xi)} \int_{X(a_0,a_1)} \mathrm{Res}_{s=0}(\cdots),$$

where $S_0(y,\xi)$ is the set of effective classes satisfying a linear condition [**?**]. The key structural features are:

1. The difference across the wall is a *sum over effective classes*.
2. Each class contributes an integral over a moduli space $X(a_0, a_1)$.
3. The contribution is computed by a residue.

16.4.2 Proposed wall-crossing formula for the tower

[Wall-crossing formula] Prove that for a path ν_t in the moduli space $\mathcal{M}_n$ crossing the image of the discriminant locus Δ transversally at a point ν_0, the change in the fibre $F_n(\nu)$ is given by

$$\Delta F_n(\nu_0) = \sum_{\sigma\in\Sigma(\nu_0)} c_\sigma \cdot F_\sigma(\nu_0),$$

where:

1. $\Sigma(\nu_0)$ is the set of degenerate strata of X_n whose image in $\mathcal{M}_n$ contains ν_0.
2. $F_\sigma(\nu_0)$ is the fibre of β_n restricted to the stratum σ over ν_0.
3. c_σ is a combinatorial coefficient depending on the multiplicity of the degeneracy and the local geometry of the discriminant at ν_0.

Remark 16.1 (Status). The formula is proposed as the algebraic-statistics analogue of the Donaldson wall-crossing formula. It is not established. The coefficient c_σ must be determined by the local structure of the defectivity locus, and the sum must be shown to converge (in the finite case, it is a finite sum).

16.4.3 Example: a single component collision

Consider the family

$$\nu_\varepsilon = p\,\mathcal{N}(\mu,\Sigma) + (1-p)\,\mathcal{N}(\mu+\varepsilon v,\Sigma), \qquad 0<p<1,\ \Sigma\succ 0,\ v\neq 0.$$

For $\varepsilon \neq 0$, the two components are distinct and the fibre is a singleton in the minimal representation. At $\varepsilon = 0$, the two components coincide and the fibre becomes a continuous family

$$F_1(\nu_0) = \{(p_1,p_2;\mu,\Sigma) : p_1+p_2=1,\ p_1,p_2\geq 0\},$$

parameterized by the split of the total weight between the two coincident components.

The wall-crossing formula in this case is

$$\Delta F_1(\nu_0) = F_{\text{collision}}(\nu_0),$$

where $F_{\text{collision}}(\nu_0)$ is the continuous family parameterized by the weight split. The coefficient $c_\sigma = 1$ in this case.

16.5 The Berkovich lift

The wall-crossing formula of Obligation 16.4.2 is an algebraic statement about the fibres of β_n. The Berkovich bridge of Section 15 proposes to lift this to a cohomological statement on the Berkovich space $\Phi(X_n)$.

[Berkovich wall-crossing] Construct the Čech complex of the G-topology $\mathcal{T}_{\Phi(\mathcal{M}_n)}$ on the Berkovich moduli space, and define a map from wall-crossing data (the coefficient c_σ and the stratum F_σ) to Čech 1-cocycles. Prove that the wall-crossing formula of Obligation 16.4.2 is equivalent to the vanishing of the corresponding Čech cocycle.

Remark 16.2 (Status). This obligation connects the fibre document, the Berkovich bridge, and the double-category paper. It is the central open problem of the proposed work. It requires:

1. The Čech complex of the G-topology (Obligation 15.6 of Section 15).
2. A map from wall-crossing data to cocycles.
3. A proof that the wall-crossing formula is equivalent to the cocycle vanishing.

None of these is currently established.

16.6 The finite test case

[Finite wall-crossing computation] For $B = \{0, \dots, n-1\}$ and $A = \mathbb{R}^B$, construct the secant variety stratification of the finite Gaussian mixture tower explicitly. Compute the wall-crossing formula for a simple component collision. Verify that the resulting Čech cocycle is a coboundary if and only if the collision is generic.

This is a finite computation. The moment variety $\mathcal{G}_{d,2}$ for finite d is computable by Gröbner bases. The secant variety $\text{Sec}_k(\mathcal{G}_{d,2})$ is computable by elimination. The discriminant locus is the zero set of the coincidence functions δ_{ij}. The wall-crossing formula is a finite sum.

16.7 What the proposed work establishes and what it does not

Statement	**Status**
Gaussian mixtures are secant varieties of the Gaussian moment variety	Established [**?**]
Identifiability corresponds to non-defectivity of the secant variety	Established [**?**]
The tower is a nested sequence of secant varieties	Proposed
The discriminant locus is the defectivity locus	Proposed
The wall-crossing formula (Obligation 16.4.2)	Open
The Berkovich lift (Obligation 16.5)	Open
The finite computation (Obligation 16.6)	Open

What the proposed work establishes. It identifies the algebraic-statistics framework (secant varieties, moment maps, defectivity) as the correct language for the fibre stratification of the Gaussian mixture tower. It proposes a wall-crossing formula in the style of Donaldson theory and Bridgeland stability, and it proposes to lift this formula to a Čech cocycle on the Berkovich moduli space.

What the proposed work does not establish. It does not prove the wall-crossing formula. It does not construct the Berkovich lift. It does not resolve the inheritance obligation of Section 15. It does not extend to the standard-Borel or Giry case.

Relation to mirror symmetry. The Berkovich methods used in mirror symmetry for log Calabi–Yau surfaces [**?**] are structurally parallel to the proposed Berkovich lift. Where mirror symmetry counts holomorphic cylinders, the proposed work counts hierarchical representations. Where mirror symmetry has a wall-crossing formula for broken lines, the proposed work has a wall-crossing formula for lifting obstructions. The parallel is structural, not metaphorical, but it is not yet a theorem.

16.8 Concrete next steps

1. **Compute the secant variety stratification.** For small d and k, compute the moment variety $\mathcal{G}_{d,2}$, the secant variety $\mathrm{Sec}_k(\mathcal{G}_{d,2})$, and the defectivity locus. Determine the dimension of the secant variety and the dimension of the fibres of the moment map.

2. **Prove the wall-crossing formula in the single-collision case.** For the family $\nu_\varepsilon = p\mathcal{N}(\mu, \Sigma) + (1-p)\mathcal{N}(\mu + \varepsilon v, \Sigma)$, compute $\Delta F_1(\nu_0)$ explicitly and verify the formula.

3. **Construct the Čech complex for the finite tower.** For $B = \{0, \ldots, n-1\}$, write down the G-topology on $\Phi(\mathcal{M}_n)$ and the Čech complex. Compute the obstruction cocycle for a simple wall-crossing.

4. **Connect to the double category.** Transport the wall-crossing formula to the double-category structure of Section 11. The horizontal 1-cells are the context morphisms; the vertical 1-cells are the shadows L and R; the 2-cells are the wall-crossing comparisons.

5. **State the analytic extension.** Replace the finite base with $[0, 1]$ and the finite distribution monad with the Giry monad. State the wall-crossing formula in the analytic setting and identify the obstruction to extending the finite computation.

16.9 Conclusion

The proposed work connects the fibre stratification of the Gaussian mixture tower to algebraic geometry via secant varieties of the Gaussian moment variety. The discriminant locus is the defectivity locus; the wall-crossings are defectivity jumps; the wall-crossing formula is the algebraic-statistics analogue of the Donaldson wall-crossing formula. The Berkovich bridge lifts the wall-crossing formula to a Čech cocycle on the Berkovich moduli space, connecting the fibre document to the double-category paper.

The proposal is stated as a programme with explicit obligations. The finite case is checkable. The analytic extension is open. The connection to mirror symmetry is structural. The next step is to compute the secant variety stratification and the wall-crossing formula in the single-collision case, and to construct the Čech complex in the finite test case.

17 Conclusion and open problems

The posterior tower suggests a useful distinction between two forms of pooling and motivates the study of their interaction with Beck defects. The finite conditional-expectation setting supplies a rigorous operator-level starting point: conditional-expectation projections commute when the sigma-algebras are nested, and they commute under suitable conditional-independence hypotheses. The composition of such projections must not be confused with projection onto the join of the sigma-algebras. Example 4.6 shows that the hypotheses are genuinely necessary.

The categorical construction remains a separate task. To obtain a double category, the context and measure layers must be connected by explicitly defined functors or lifted shadow maps. The lifting functor F has been constructed (Definition 5.5) and shown to be compatible with composition. The double category of 2-cells in **2Cat** has been constructed (Definition 11.1), and its interchange law is automatic (Theorem 11.2). The Gaussian instance verifies the interchange law for the shadows under conditional independence and provides a candidate level-one boundary for a double-category square.

The principal open problems are:

1. Define measurable shadow labels, partitions, and conditional-barycenter operators for the intended posterior tower.
2. Prove that the measure-valued shadows inherit the relevant commuting conditional-expectation identity.
3. Construct the comparison maps for the Beck defects and prove their naturality and coherence, using the lifting functor F of Definition 5.5 and the (L, R)-compatible subcategory of Definition 5.10.
4. Prove, rather than assume, any factorization of the Beck defects through L, R, or their pair.
5. Construct shadow transports and verify their naturality and all Beck axioms for Option A or Option B.
6. Specify a suitable measurable category for the Giry extension and prove the required closure, measurability, and barycenter properties.
7. Determine whether a nontrivial higher-categorical structure is needed after the double-category construction is completed.
8. Build finite examples that distinguish nested, conditionally independent, and noncommuting regimes.

The appropriate current conclusion is therefore a conditional operator theorem, a double-category construction with automatic interchange, and a categorical program for the comparison maps.

Code and repository

The original project repository cited for the computational checks is:

https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/papers

The precise scripts and frozen outputs used for any theorem-specific check should be identified in the corresponding release or appendix.

References

[1] S. Abramsky and A. Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011.

[2] S. Abramsky, S. Mansfield, and R. S. Barbosa. The cohomology of non-locality and contextuality. *Electronic Proceedings in Theoretical Computer Science*, 95:1–14, 2012.

[3] S. Amari and H. Nagaoka. *Methods of Information Geometry*. American Mathematical Society, 2000.

[4] V. K. Awasthi. Bands: perturbation, tearing and breaking of coupling structures. Technical report, Johns Hopkins University, October 2026.

[5] V. K. Awasthi. Bridges: from probability and entropy to branes. Technical note, October 2026.

[6] V. K. Awasthi. Conditionals, couplings and toric geometry. Technical report, October 2026.

[7] V. K. Awasthi. The three-level posterior tower. Technical report, October 2026.

[8] V. K. Awasthi. Three layers of realisation. Technical report, 2026.

[9] V. K. Awasthi. Empirical site separation. Technical report, October 2026.

[10] J. Beck. Distributive laws. In *Seminar on Triples and Categorical Homology Theory*, Lecture Notes in Mathematics 80, pages 119–140. Springer, 1969.

[11] G. Böhm. The weak theory of monads. *Advances in Mathematics*, 225(1):1–32, 2010.

[12] G. Böhm, S. Lack, and R. Street. On the 2-categories of weak distributive laws. *Communications in Algebra*, 39(12):4567–4583, 2011.

[13] O. Calin and C. Udrişte. *Geometric Modeling in Probability and Statistics*. Springer, 2014.

[14] E. Cheng. Distributive laws for Lawvere theories. In *Proceedings of CALCO 2021*, LIPIcs 211, 2021.

[15] E. N. Dzhafarov and J. V. Kujala. Context-content systems of random variables. *Journal of Mathematical Psychology*, 74:11–33, 2016.

[16] E. N. Dzhafarov, V. H. Cervantes, and J. V. Kujala. Contextuality in canonical systems. *Philosophical Transactions of the Royal Society A*, 375:20160389, 2017.

[17] T. Fritz. A synthetic approach to Markov kernels. *Advances in Mathematics*, 370:107239, 2020.

[18] M. Giry. A categorical approach to probability theory. In *Categorical Aspects of Topology and Analysis*, Lecture Notes in Mathematics 915, pages 68–85. Springer, 1982.

[19] P. G. Goerss and J. F. Jardine. *Simplicial Homotopy Theory*. Birkhäuser, 2009.

[20] P. T. Johnstone. *Sketches of an Elephant*, volume 1. Oxford University Press, 2002.

[21] S. Lack. Codescent objects and coherence. *Journal of Pure and Applied Algebra*, 175(1–3):223–241, 2002.

[22] S. Mac Lane. *Categories for the Working Mathematician*. Graduate Texts in Mathematics 5, Springer, 2nd edition, 1998.

[23] S. Mac Lane and I. Moerdijk. *Sheaves in Geometry and Logic*. Springer, 1992.

[24] E. Moggi. Notions of computation and monads. *Information and Computation*, 93(1):55–92, 1991.

[25] J. Power and H. Watanabe. Combining a monad and a comonad. *Theoretical Computer Science*, 280(1–2):137–162, 2002.

[26] E. Riehl. *Category Theory in Context*. Dover, 2016.

[27] R. Street. The formal theory of monads. *Journal of Pure and Applied Algebra*, 2(2):149–168, 1972.

[28] R. Street. Weak distributive laws. *Theory and Applications of Categories*, 22(12):313–320, 2009.

[29] J.-P. Vigneaux. *Topology of Statistical Systems*. Ph.D. thesis, Université Paris Diderot, 2019.

[30] C. Villani. *Optimal Transport: Old and New*. Springer, 2009.

[31] S. Zhang, L. Guo, and W. Keigher. Monads and distributive laws for Rota–Baxter algebras. *Advances in Applied Mathematics*, 72:139–165, 2016.